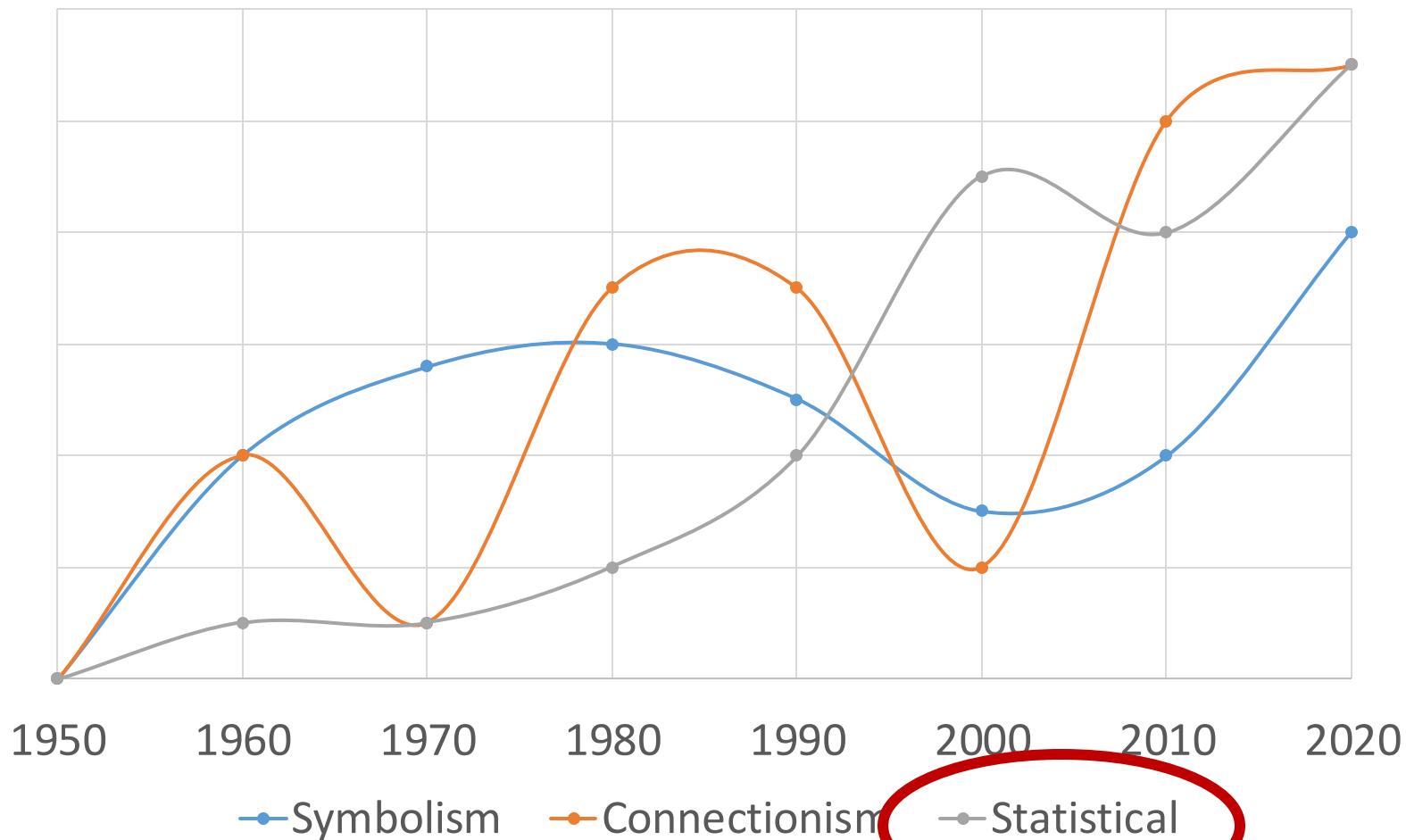


Three types of (strong) AI approaches



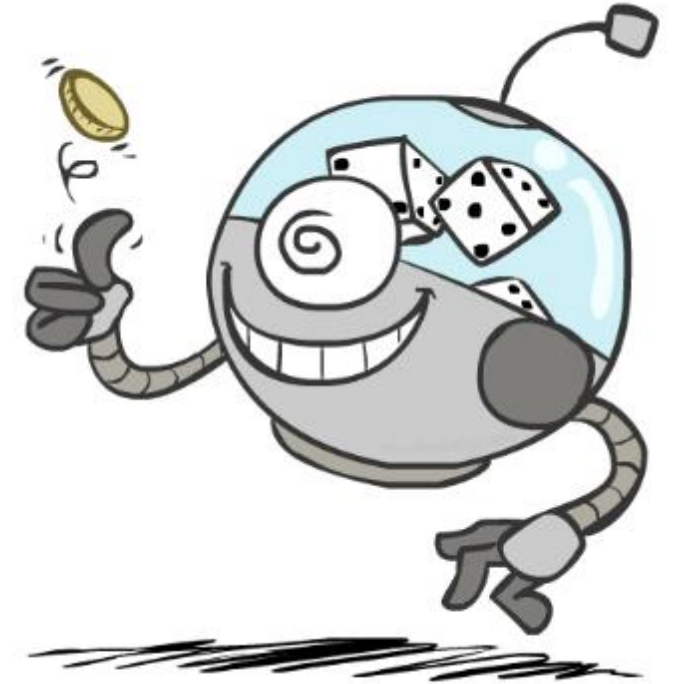
Probability



AIMA Chapter 13

Outline

- Probability
 - Random Variables
 - Joint and Marginal Distributions
 - Conditional Distributions
 - Inference
 - Product Rule, Chain Rule, Bayes' Rule



Uncertainty

- My flight to New York is scheduled to leave at 11:25
 - Let action A_t = leave home t minutes before flight and drive to the airport
 - Will A_t ensure that I catch the plane?
- Problems:
 - noisy sensors (radio traffic reports, Google maps)
 - uncertain action outcome (car breaking down, accident, etc.)
 - partial observability (other drivers' plans, etc.)
 - immense complexity of modelling and predicting traffic, security line, etc.

Responses to uncertainty

- Ignore it – map directly from percept stream (known) to actions
 - Hopeless!
- Some sort of softening of logical rules (*fudge factors*)
 - $A_{1440} \rightarrow_{0.9999} \text{CatchPlane}$
 - $\text{CatchPlane} \rightarrow_{0.95} \neg \text{MajorTrafficJam}$
 - Hence, chaining these together, $A_{1440} \rightarrow_{0.949} \neg \text{MajorTrafficJam}$
 - Oops
- Probability (Mahaviracarya (9th C.), Cardamo (1565))
 - Given the available evidence and the choice A_{120} , I will catch the plane with probability 0.92

Probability

- Probability

- Given the available evidence and the choice A_{120} , I will catch the plane with probability 0.92

- **Subjective** or **Bayesian** probability:

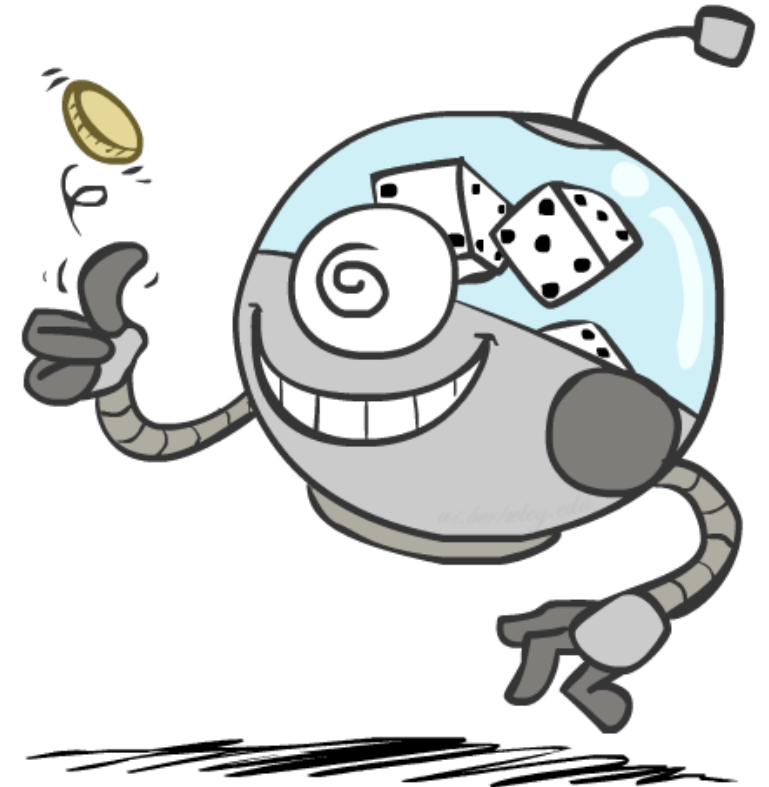
- Probabilities relate propositions to one's own state of knowledge
 - ignorance: lack of relevant facts, initial conditions, etc.
 - laziness: failure to list all exceptions, compute detailed predictions, etc.
- Not claiming a “probabilistic tendency” in the actual situation (traffic is not like quantum mechanics)

Decisions

- Suppose I believe
 - $P(\text{CatchPlane} \mid A_{60}, \text{all my evidence...}) = 0.51$
 - $P(\text{CatchPlane} \mid A_{120}, \text{all my evidence...}) = 0.97$
 - $P(\text{CatchPlane} \mid A_{1440}, \text{all my evidence...}) = 0.9999$
- Which action should I choose?
- Depends on my **preferences** for, e.g., missing flight, airport food, etc.
- **Utility theory** is used to represent and infer preferences
- **Decision theory** = utility theory + probability theory
- **Maximize expected utility** : $a^* = \operatorname{argmax}_a \sum_s P(s \mid a) U(s)$

Random Variables

- A random variable is some aspect of the world about which we (may) have uncertainty
 - R = Is it raining?
 - T = Is it hot or cold?
 - D = How long will it take to drive to work?
 - L = Where is the pacman?
- We denote random variables with capital letters
- Like variables in a CSP, random variables have domains
 - R in $\{\text{true}, \text{false}\}$ (often write as $\{+r, -r\}$)
 - T in $\{\text{hot}, \text{cold}\}$
 - D in $[0, \infty)$
 - L in possible locations, maybe $\{(0,0), (0,1), \dots\}$



Probability Distributions

- Associate a probability with each value of a random variable

- Temperature:



$P(T)$

T	P
hot	0.5
cold	0.5

- Weather:



$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

- A probability is a single number

$$P(W = \text{rain}) = 0.1 \quad \text{Shorthand notation: } P(\text{rain}) = P(W = \text{rain}),$$

- Must have: $\forall x \ P(X = x) \geq 0$ and $\sum_x P(X = x) = 1$

Joint Distributions

- A *joint distribution* over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey: $P(x_1, x_2, \dots, x_n) \geq 0$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- Size of distribution for n variables with domain size d? d^n
 - For all but the smallest distributions, cannot write out by hand!

Probabilistic Models

- A probabilistic model is a joint distribution over a set of random variables
- Probabilistic models:
 - (Random) variables with domains
 - Joint distributions: say whether assignments (outcomes) are likely
 - Ideally: only certain variables directly interact
- Constraint satisfaction problems:
 - Variables with domains
 - Constraints: state whether assignments are possible
 - Ideally: only certain variables directly interact

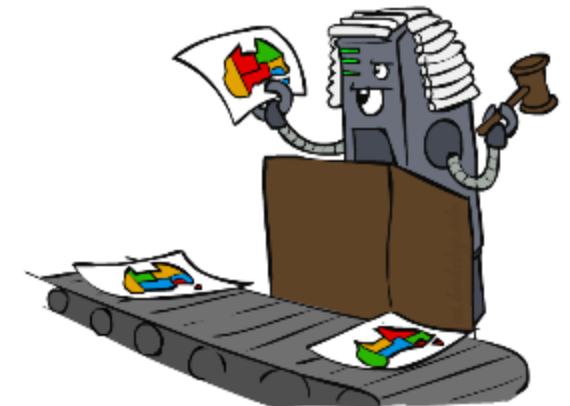
Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



Constraint over T,W

T	W	P
hot	sun	T
hot	rain	F
cold	sun	F
cold	rain	T



Probabilities of events

- An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

- Given a joint distribution over all variables, we can compute any event probability!
 - Probability that it's hot AND sunny?
 - Probability that it's hot?
 - Probability that it's hot OR sunny?

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(t) = \sum_w P(t, w)$$

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4

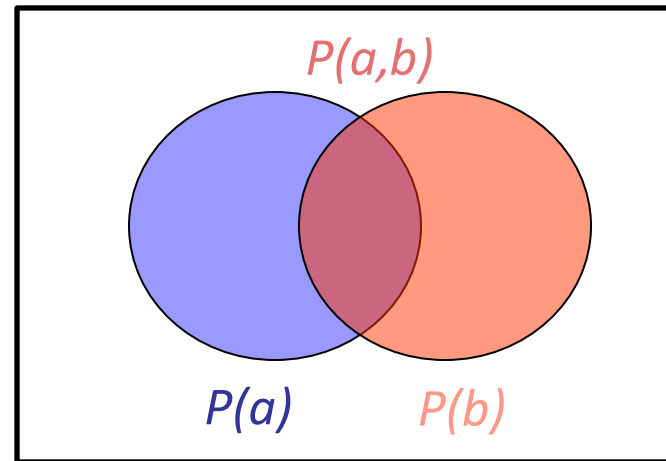
$$P(w) = \sum_t P(t, w)$$

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Conditional Probabilities

- The probability of an event given that another event has occurred

$$P(a|b) = \frac{P(a,b)}{P(b)}$$



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = 0.4$$

$$\begin{aligned} &= P(W = s, T = c) + P(W = r, T = c) \\ &= 0.2 + 0.3 = 0.5 \end{aligned}$$

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Joint Distribution

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Conditional Distributions

$$P(W|T)$$

$$P(W|T = \text{hot})$$

W	P
sun	0.8
rain	0.2

$$P(W|T = \text{cold})$$

W	P
sun	0.4
rain	0.6

Normalization Trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\begin{aligned}P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\&= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\&= \frac{0.2}{0.2 + 0.3} = 0.4\end{aligned}$$



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

$$\begin{aligned}P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\&= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\&= \frac{0.3}{0.2 + 0.3} = 0.6\end{aligned}$$

Normalization Trick

$$\begin{aligned} P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\ &= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\ &= \frac{0.2}{0.2 + 0.3} = 0.4 \end{aligned}$$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence



$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

$$\begin{aligned} P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\ &= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\ &= \frac{0.3}{0.2 + 0.3} = 0.6 \end{aligned}$$

Normalization Trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence



$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

- Why does this work? Sum of selection is $P(\text{evidence})$! ($P(T=c)$, here)

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$

Probabilistic Inference

- Probabilistic inference
 - compute a desired probability from other known probabilities (e.g. conditional from joint)
- We generally compute conditional probabilities
 - These represent the agent's beliefs given the evidence
 - $P(\text{on time} \mid \text{no reported accidents}) = 0.90$



Inference by Enumeration

General case:

- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $\left. \begin{array}{l} X_1, X_2, \dots, X_n \\ \text{All variables} \end{array} \right\}$

We want:

$$P(Q|e_1 \dots e_k)$$

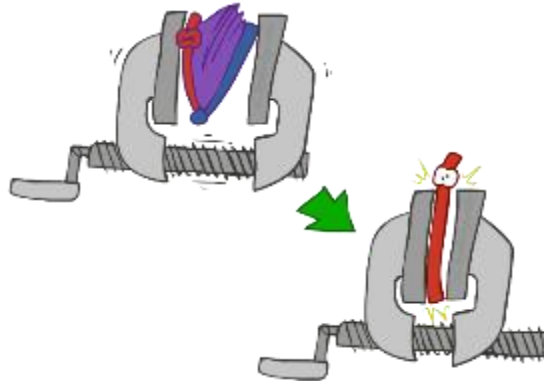
- Step 1: Select the entries consistent with the evidence



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

2 0.15

- Step 2: Sum out H to get joint of Query and evidence



- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots, X_n}$$

Inference by Enumeration

1. Select the entries consistent with the evidence
2. Sum out H to get joint of Query and evidence
3. Normalize

- $P(W \mid \text{winter})?$ sun: 0.5, rain: 0.5
- $P(W \mid \text{winter, hot})?$ sun: 0.67, rain: 0.33

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

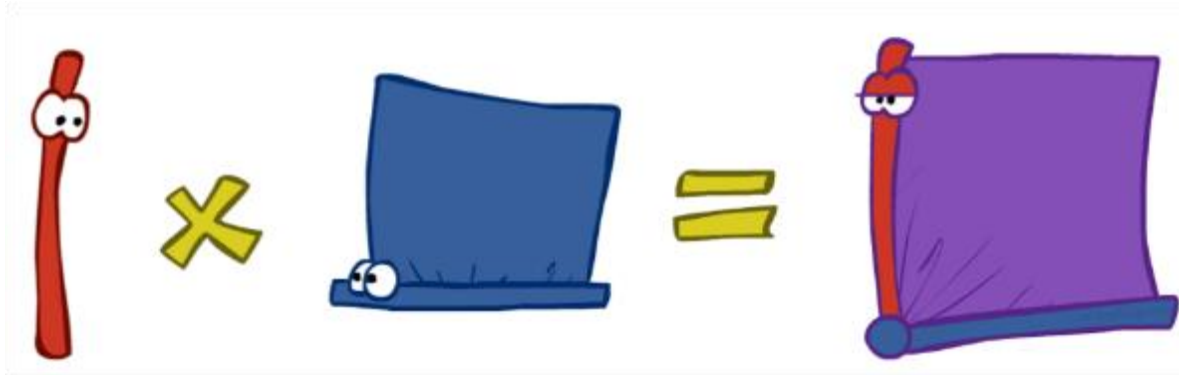
Inference by Enumeration

- Obvious problems:
 - Worst-case time complexity $O(d^n)$
 - Space complexity $O(d^n)$ to store the joint distribution

The Product Rule

- Sometimes have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y) \iff P(x|y) = \frac{P(x, y)}{P(y)}$$



The Product Rule

$$P(y)P(x|y) = P(x, y)$$

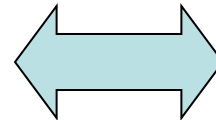
- Example:

$P(W)$

W	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3



$P(D, W)$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

The Chain Rule

- More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

Bayes' Rule

- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Why is this at all helpful?
 - Lets us build one conditional from its reverse
 - Often one conditional is tricky but the other one is simple
 - Foundation of many systems we'll see later
- In the running for most important AI equation!

That's my rule!



Inference with Bayes' Rule

- Example: Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

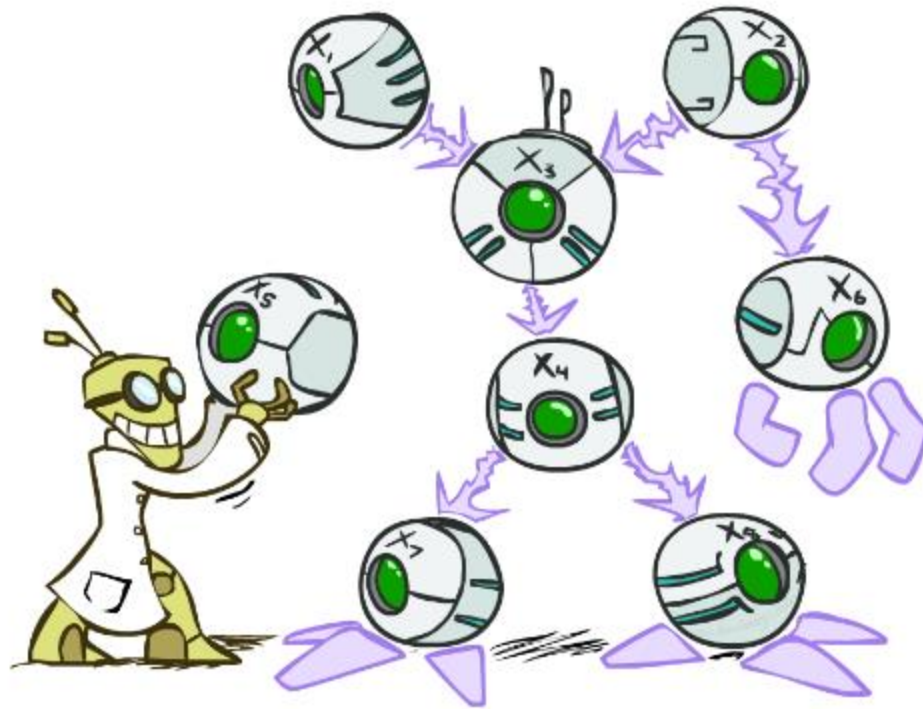
- Example:

- M: meningitis, S: stiff neck

$$\left. \begin{aligned} P(+m) &= 0.0001 \\ P(+s|+m) &= 0.8 \\ P(+s|-m) &= 0.01 \end{aligned} \right\} \begin{array}{l} \text{Example} \\ \text{gives} \end{array}$$

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{P(+s|+m)P(+m)}{P(+s|+m)P(+m) + P(+s|-m)P(-m)} = \frac{0.8 \times 0.0001}{0.8 \times 0.0001 + 0.01 \times 0.9999}$$

Bayesian Networks



AIMA Chapter 14.1, 14.2

Additional Reference

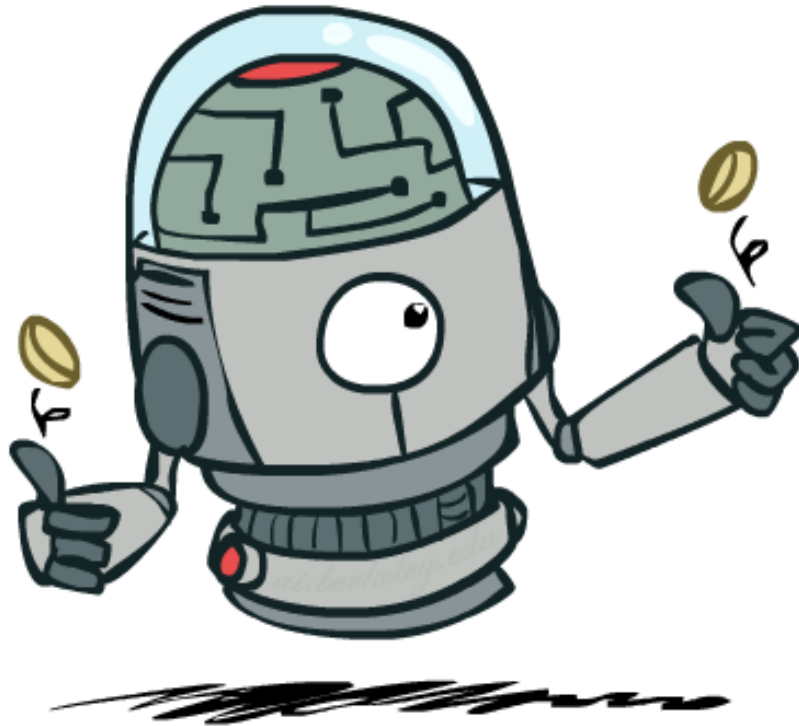
- [PRML] Pattern Recognition and Machine Learning, Christopher Bishop, Springer 2006.
 - Chapter 8.1 - 8.3

Probabilistic Models

- What do we do with probabilistic models?
 - We (or our agents) need to reason about unknown variables, given evidence
 - Example: explanation (diagnostic reasoning)
 - Example: prediction (causal reasoning)
 - Example: making decisions based on expected utility
- How do we build models, avoiding the d^n blowup?



Independence



Independence

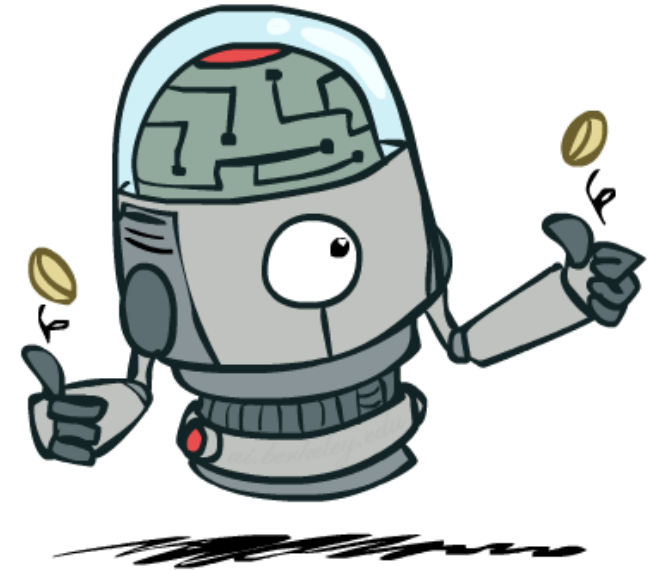
- Two variables X and Y are (absolutely) **independent** if

$$\forall x, y \quad P(x, y) = P(x)P(y)$$

- This says that their joint distribution **factors** into a product of two simpler distributions
- Combine with product rule $P(x, y) = P(x|y)P(y)$ we obtain another form:

$$\forall x, y \quad P(x|y) = P(x) \quad \text{or} \quad \forall x, y \quad P(y|x) = P(y)$$

- Example: two dice rolls $Roll_1$ and $Roll_2$
 - $P(Roll_1=5, Roll_2=5) = P(Roll_1=5)P(Roll_2=5) = 1/6 \times 1/6 = 1/36$
 - $P(Roll_2=5 \mid Roll_1=5) = P(Roll_2=5)$



Conditional Independence

- Unconditional (absolute) independence is rare
- *Conditional independence* is our most basic and robust form of knowledge about uncertain environments.
- X is conditionally independent of Y given Z $X \perp\!\!\!\perp Y | Z$

if and only if:

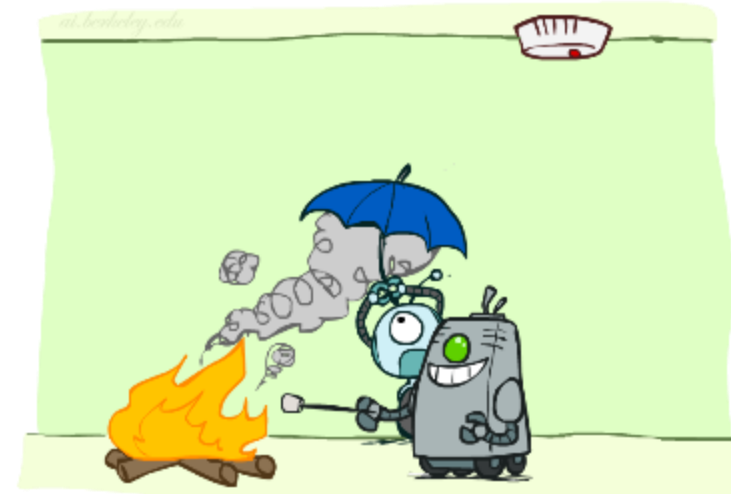
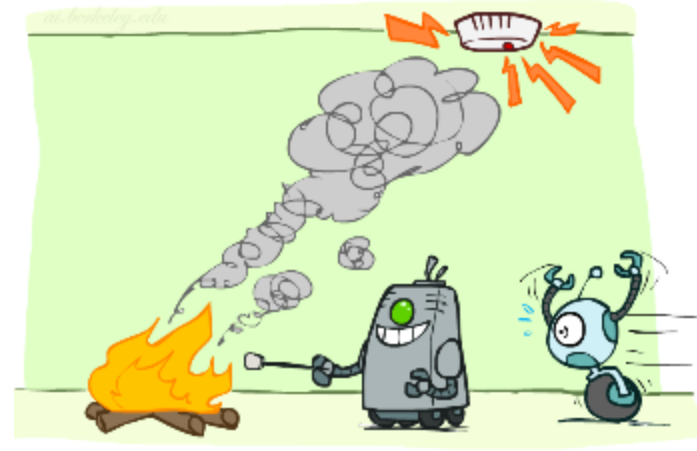
$$\forall x, y, z \quad P(x \mid y, z) = P(x \mid z)$$

or, equivalently, if and only if

$$\forall x, y, z \quad P(x, y \mid z) = P(x \mid z)P(y \mid z)$$

Conditional Independence

- What about this domain:
 - Fire
 - Smoke
 - Alarm (smoke detector)



Conditional Independence

- What about this domain:
 - Traffic
 - Umbrella
 - Raining



Conditional Independence and the Chain Rule

- Chain rule: $P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots$

- Trivial decomposition:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

- With assumption of conditional independence:

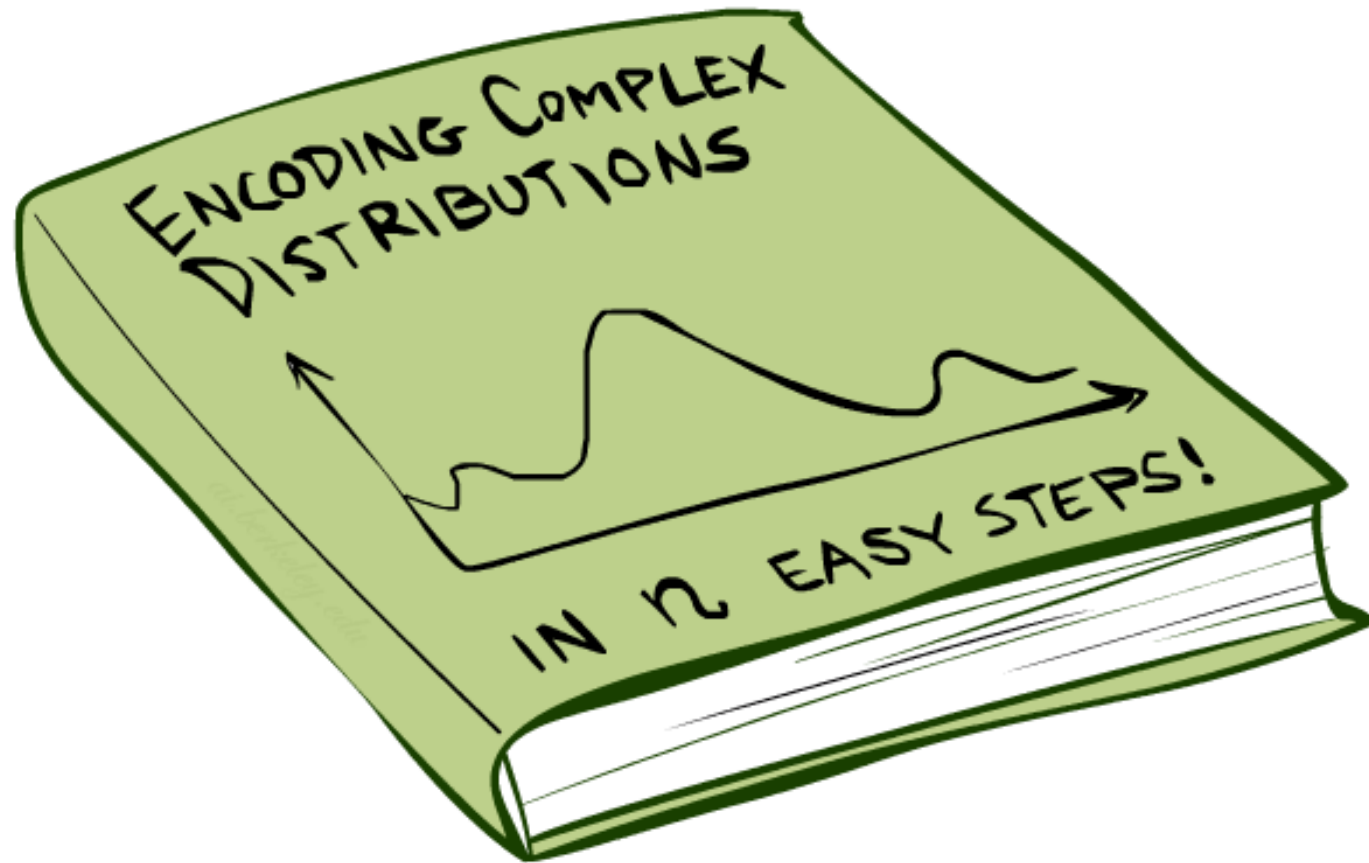
$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

Requires less space to encode!

- BayesNets / graphical models help us express conditional independence assumptions

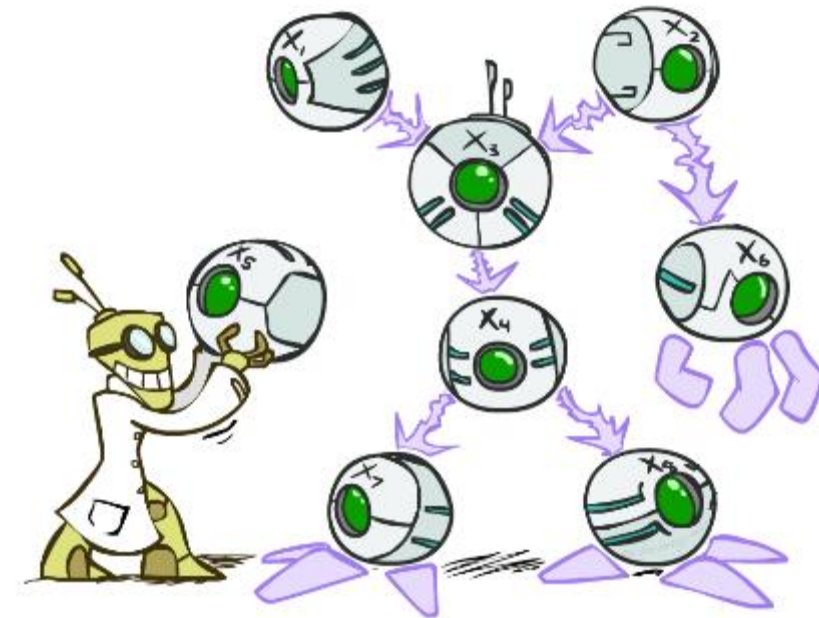


Bayesian Networks: Big Picture

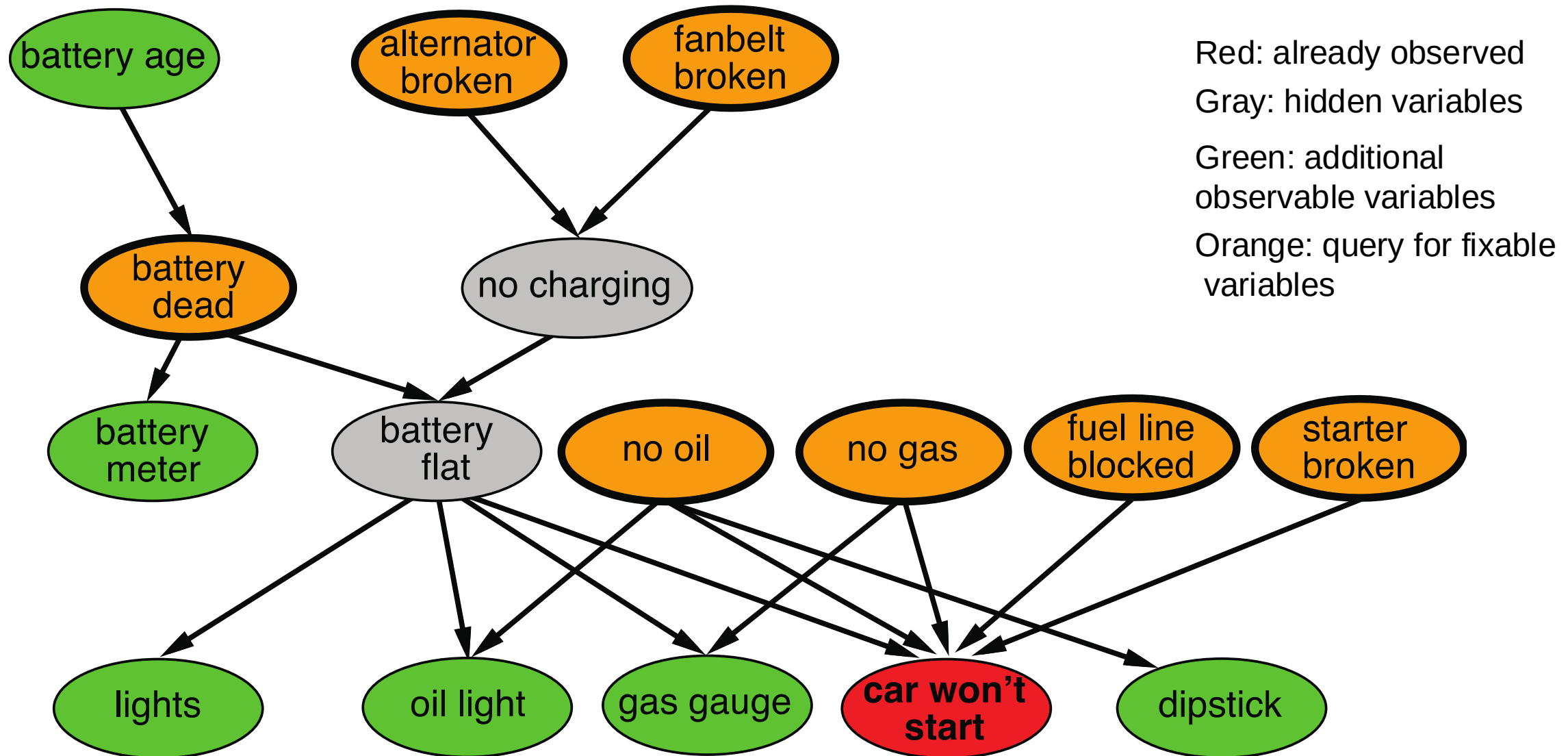


Bayesian Networks: Big Picture

- Full joint distribution tables answer every question, but:
 - Size is exponential in the number of variables
 - Need gazillions of examples to learn the probabilities
 - Inference by enumeration (summing out hidden) is too slow
- Bayesian networks:
 - Express all the conditional independence relationships in a domain
 - Factor the joint distribution into a product of small conditionals
 - Often reduce size from exponential to linear
 - Faster learning from fewer examples
 - Faster inference (linear time in some important cases)



Example Bayes Net: Amateur Car Mechanic



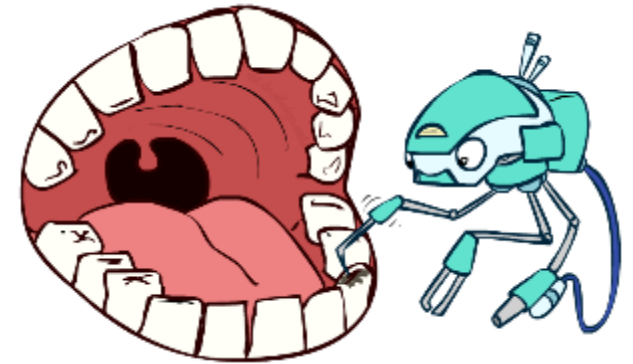
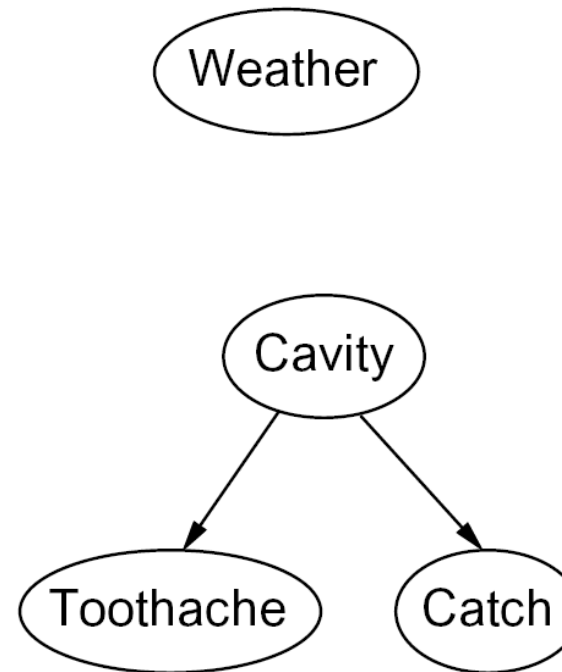
Bayesian Networks Syntax



Bayesian Networks Syntax

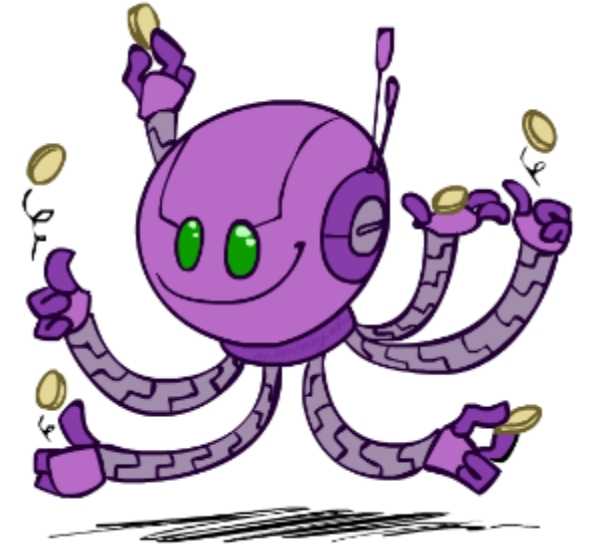


- Nodes: variables (with domains)
- Arcs: interactions
 - Indicate “direct influence” between variables
 - For now: imagine that arrows mean direct causation (in general, they may not!)
 - Formally: encode conditional independence (more later)
- No cycle is allowed!



Example: Coin Flips

- N independent coin flips



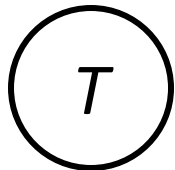
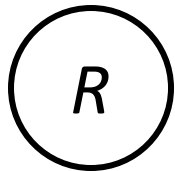
- No interactions between variables: **absolute independence**

Example: Traffic

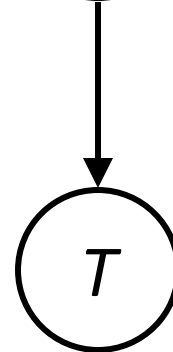
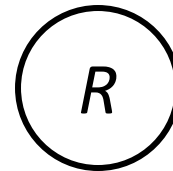
- Variables:

- R: It rains
- T: There is traffic

- Model 1: independence



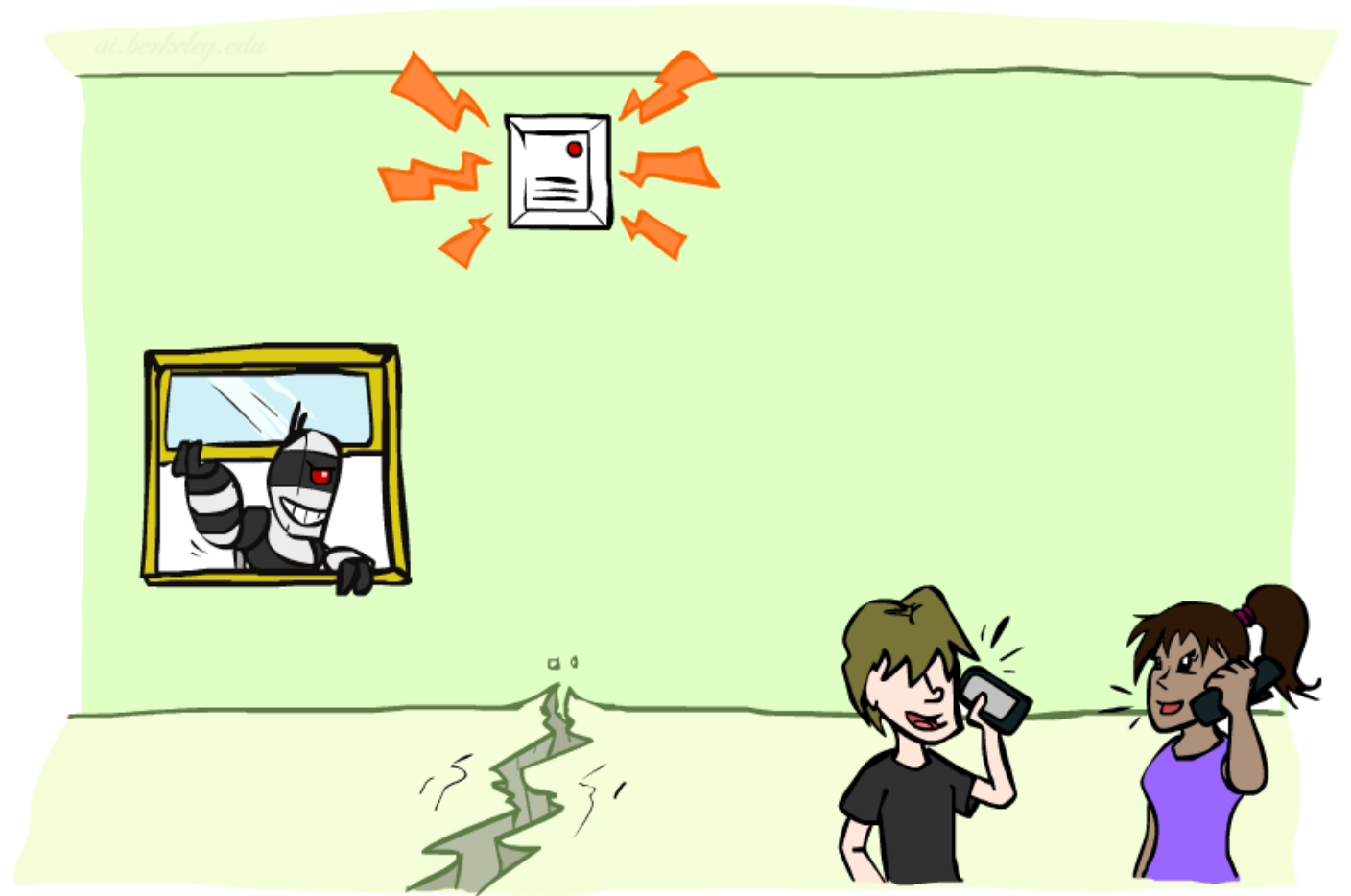
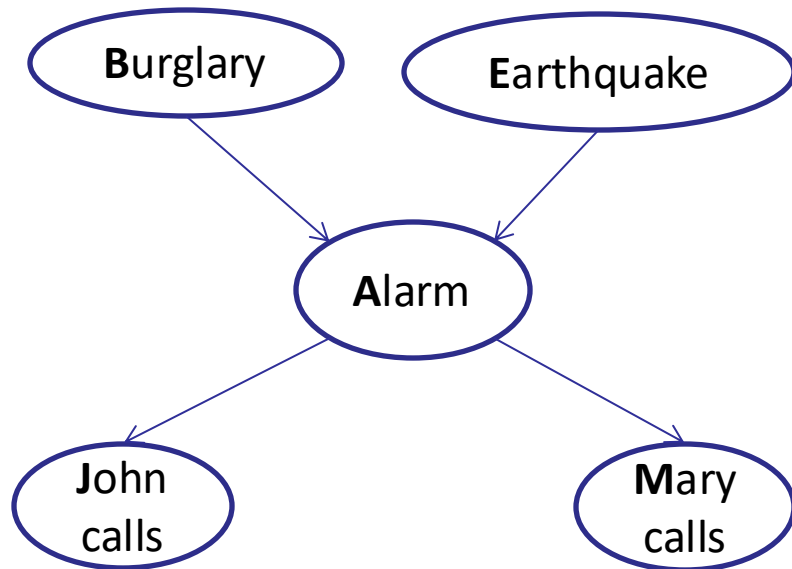
- Model 2: rain causes traffic



Example: Alarm Network

■ Variables

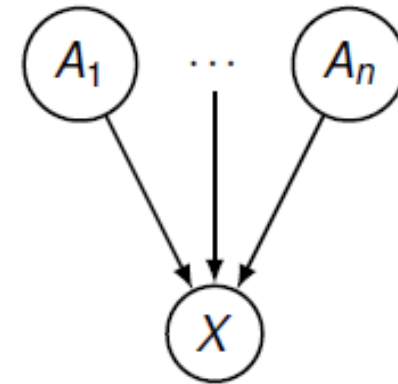
- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!



Bayesian Networks Syntax



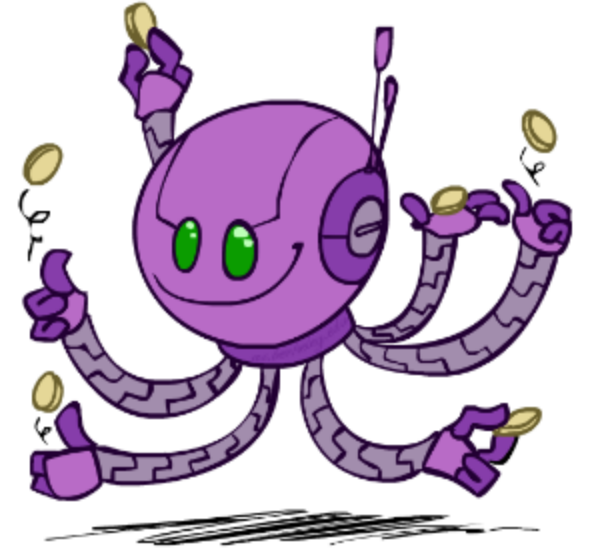
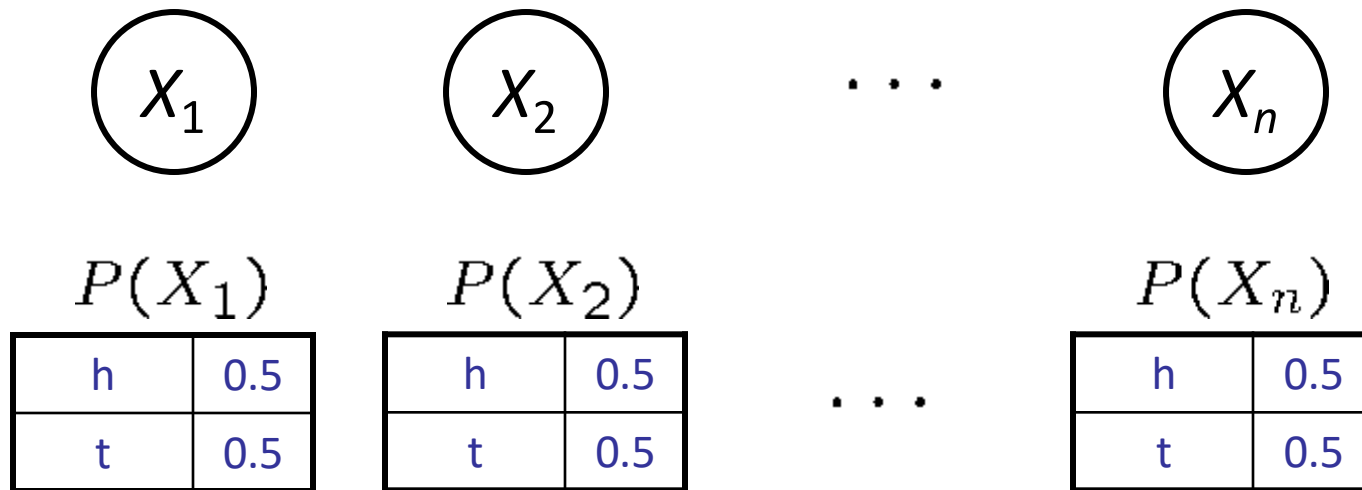
- A directed, acyclic graph
- Conditional distributions for each node given its **parent variables** in the graph
 - **CPT**: conditional probability table: each row is a distribution for child given a configuration of its parents
 - Description of a noisy “causal” process



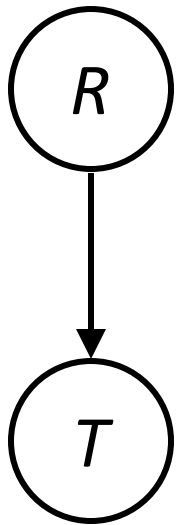
$$P(X|A_1, \dots, A_n)$$

A Bayes net = Topology (graph) + Local Conditional Probabilities

Example: Coin Flips



Example: Traffic


$$P(R)$$

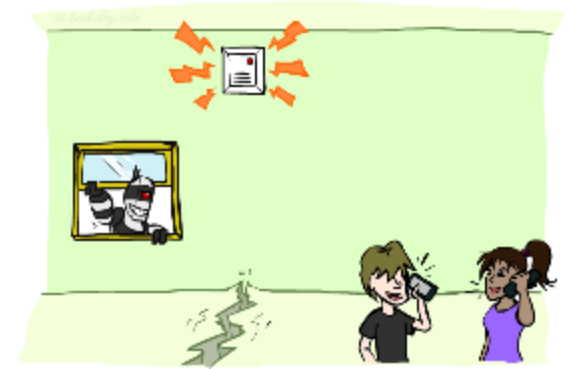
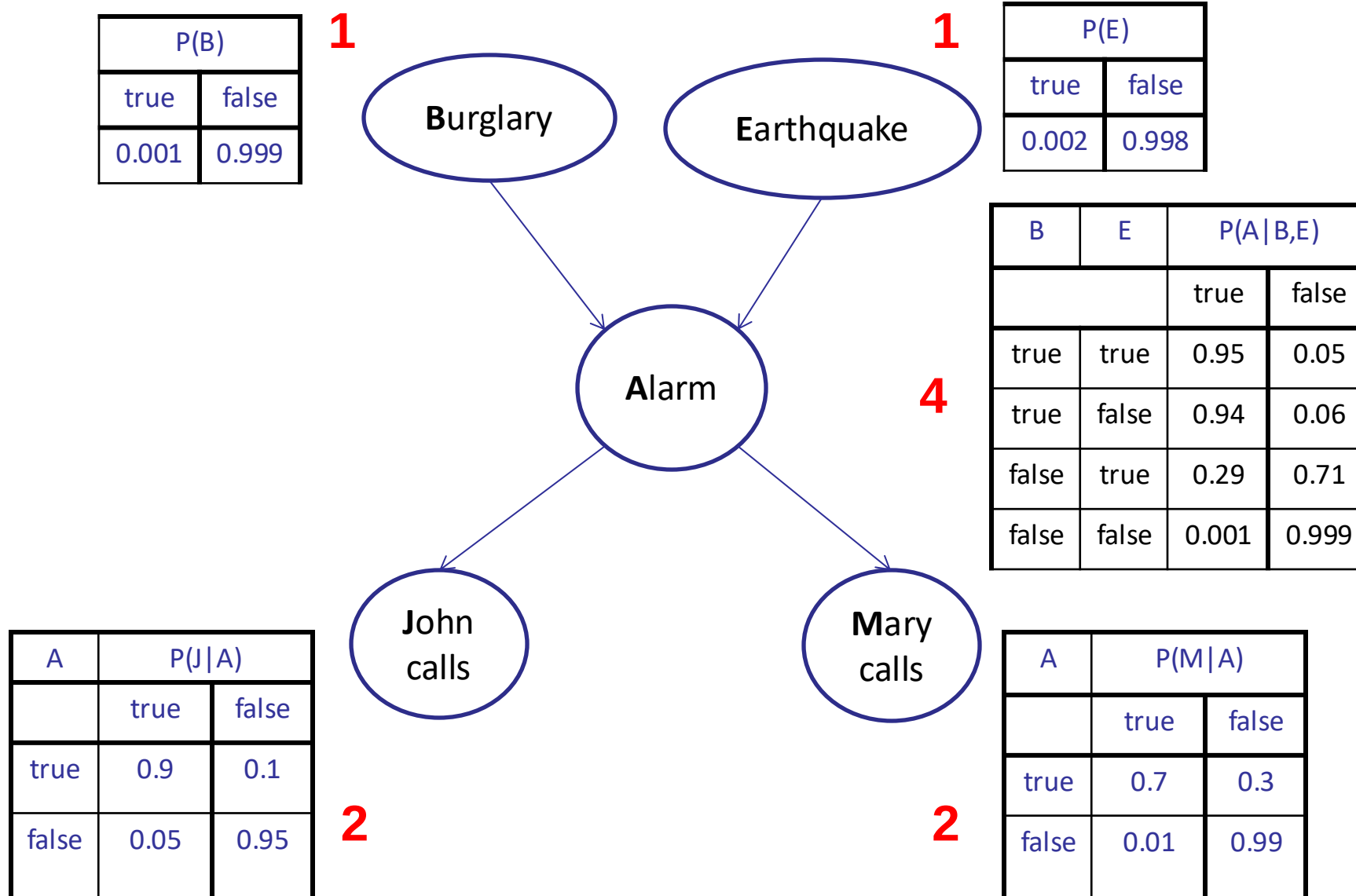
+r	1/4
-r	3/4

$$P(T|R)$$

+r	<table><tr><td>+t</td><td>3/4</td></tr><tr><td>-t</td><td>1/4</td></tr></table>	+t	3/4	-t	1/4
+t	3/4				
-t	1/4				
-r	<table><tr><td>+t</td><td>1/2</td></tr><tr><td>-t</td><td>1/2</td></tr></table>	+t	1/2	-t	1/2
+t	1/2				
-t	1/2				



Example: Alarm Network



Number of free parameters in each CPT:

- Parent domain sizes

$d_1, \dots, d_k$

- Child domain size d
- Each table row must sum to 1

$(d-1) \prod_i d_i$

General formula for sparse BNs

- Suppose
 - n variables
 - Maximum domain size is d
 - Maximum number of parents is k
- Full joint distribution has size $O(d^n)$
- Bayes net has size $O(n \cdot d^{k+1})$
 - Linear scaling with n as long as causal structure is local

Bayesian Networks Semantics



Bayesian networks global semantics



- Bayes nets encode joint distributions as product of conditional distributions on each variable:

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

Example

P(B)	
true	false
0.001	0.999

P(E)	
true	false
0.002	0.998

$$P(b, \neg e, a, \neg j, \neg m) =$$

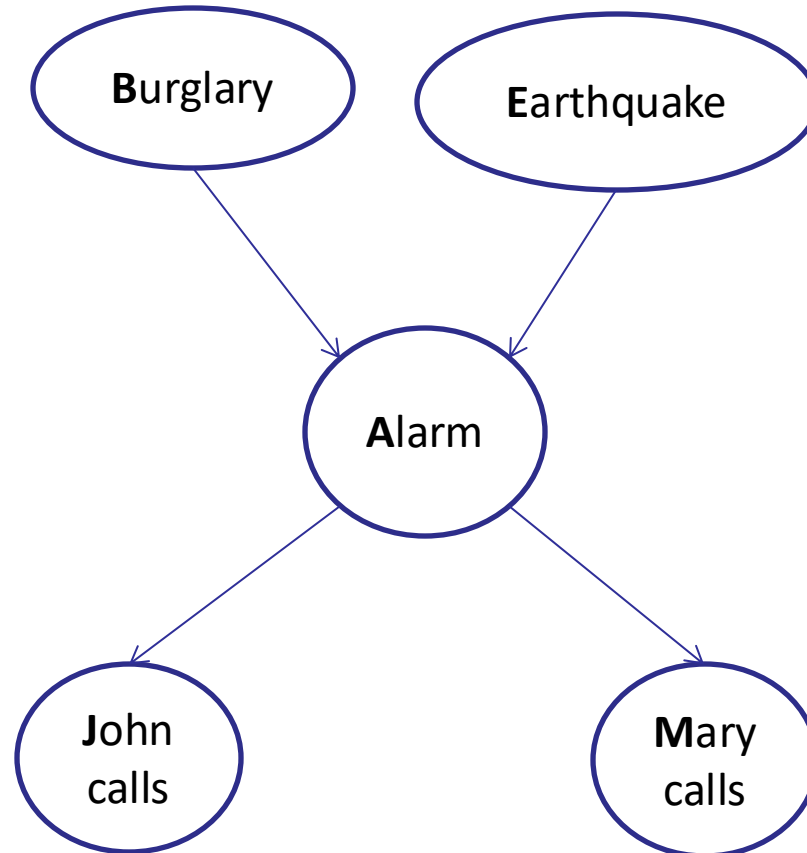
$$P(b) P(\neg e) P(a|b, \neg e) P(\neg j|a) P(\neg m|a)$$

$$=.001 \times .998 \times .94 \times .1 \times .3 = .000028$$

B	E	P(A B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

A	P(J A)	
	true	false
true	0.9	0.1
false	0.05	0.95

A	P(M A)	
	true	false
true	0.7	0.3
false	0.01	0.99



Probabilities in BNs



- Why are we guaranteed that setting

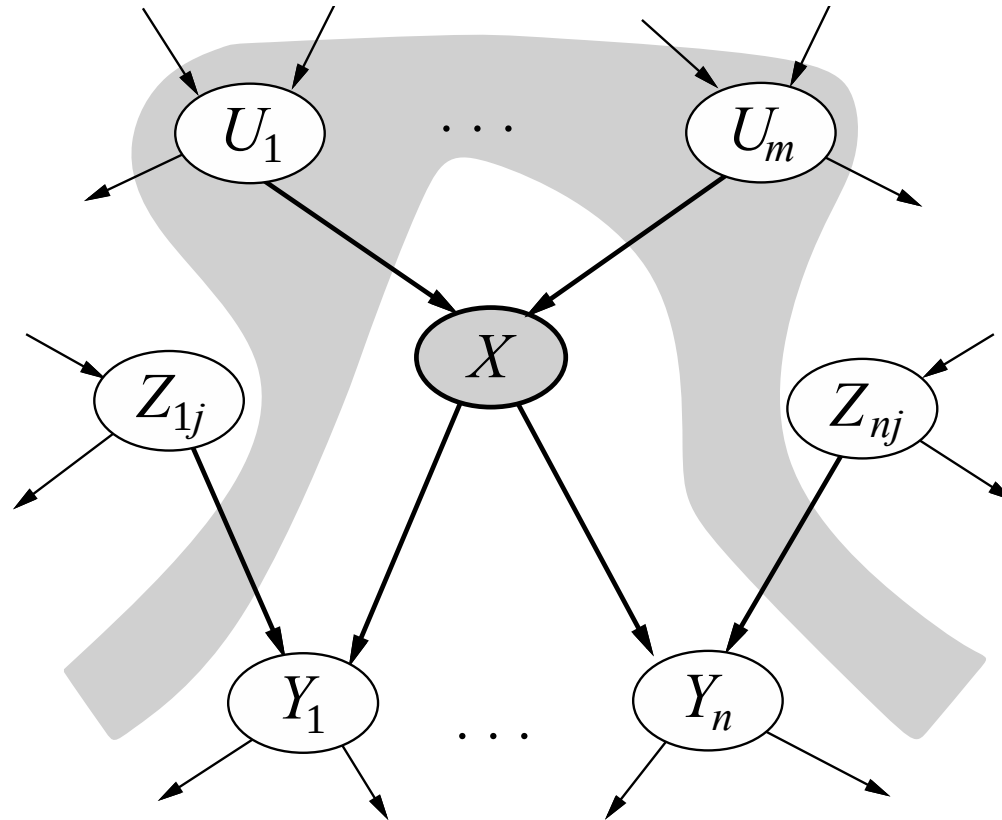
$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

results in a proper joint distribution?

- Chain rule (valid for all distributions): $P(X_1, \dots, X_n) = \prod_i P(X_i \mid X_1, \dots, X_{i-1})$
- Assume conditional independences: $P(X_i \mid X_1, \dots, X_{i-1}) = P(X_i \mid \text{Parents}(X_i))$
 - When adding node X_i , ensure parents “shield” it from other predecessors
- So the network topology implies that certain conditional independencies hold

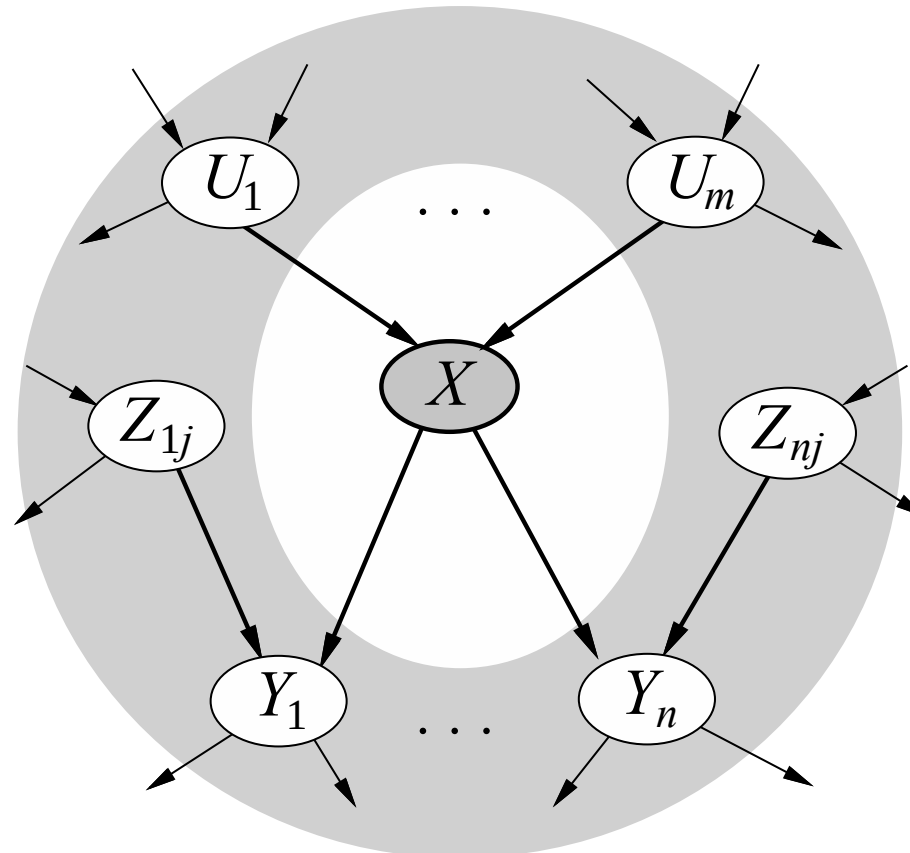
Conditional independence semantics

- *Every variable is conditionally independent of its non-descendants given its parents*
- Conditional independence semantics $\Leftrightarrow$ global semantics



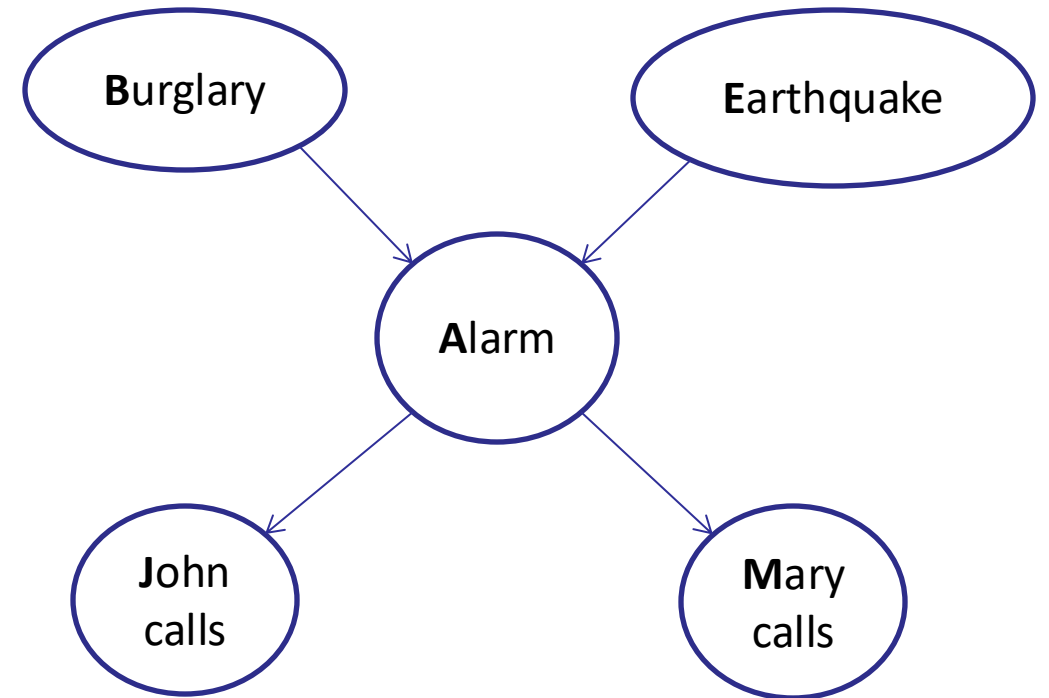
Markov blanket

- A variable's Markov blanket consists of parents, children, children's other parents
- ***Every variable is conditionally independent of all other variables given its Markov blanket***



Example

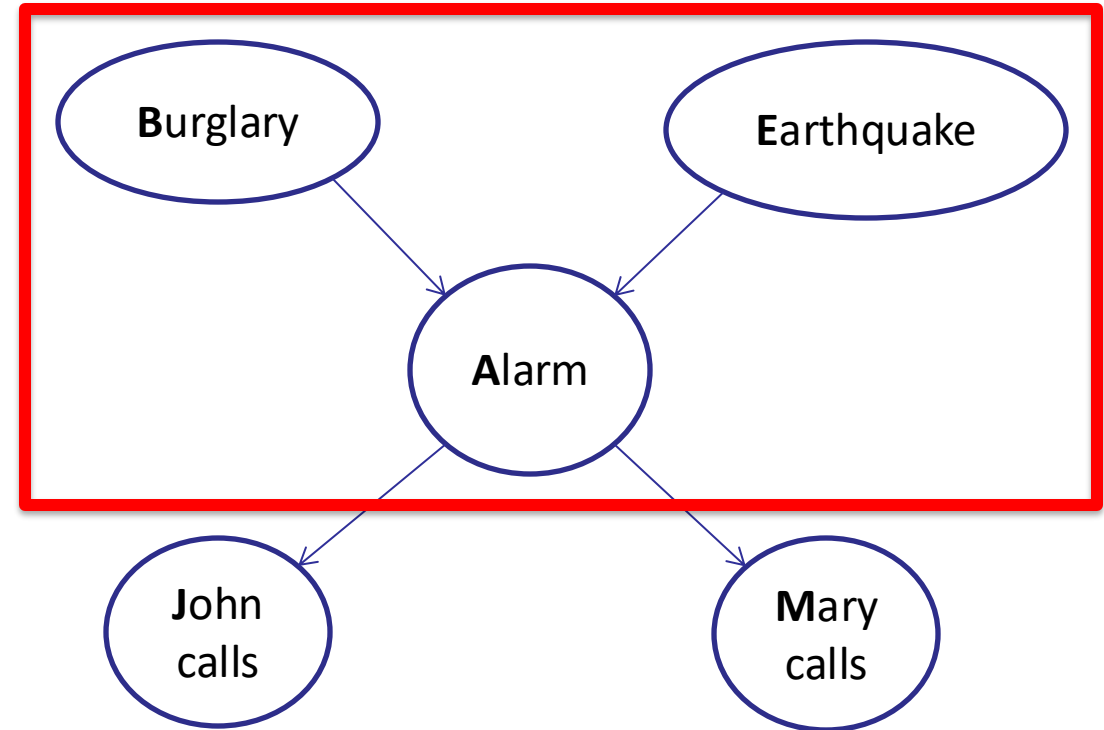
- JohnCalls independent of Burglary given Alarm?
 - Yes
- JohnCalls independent of MaryCalls given Alarm?
 - Yes
- Burglary independent of Earthquake?
 - Yes



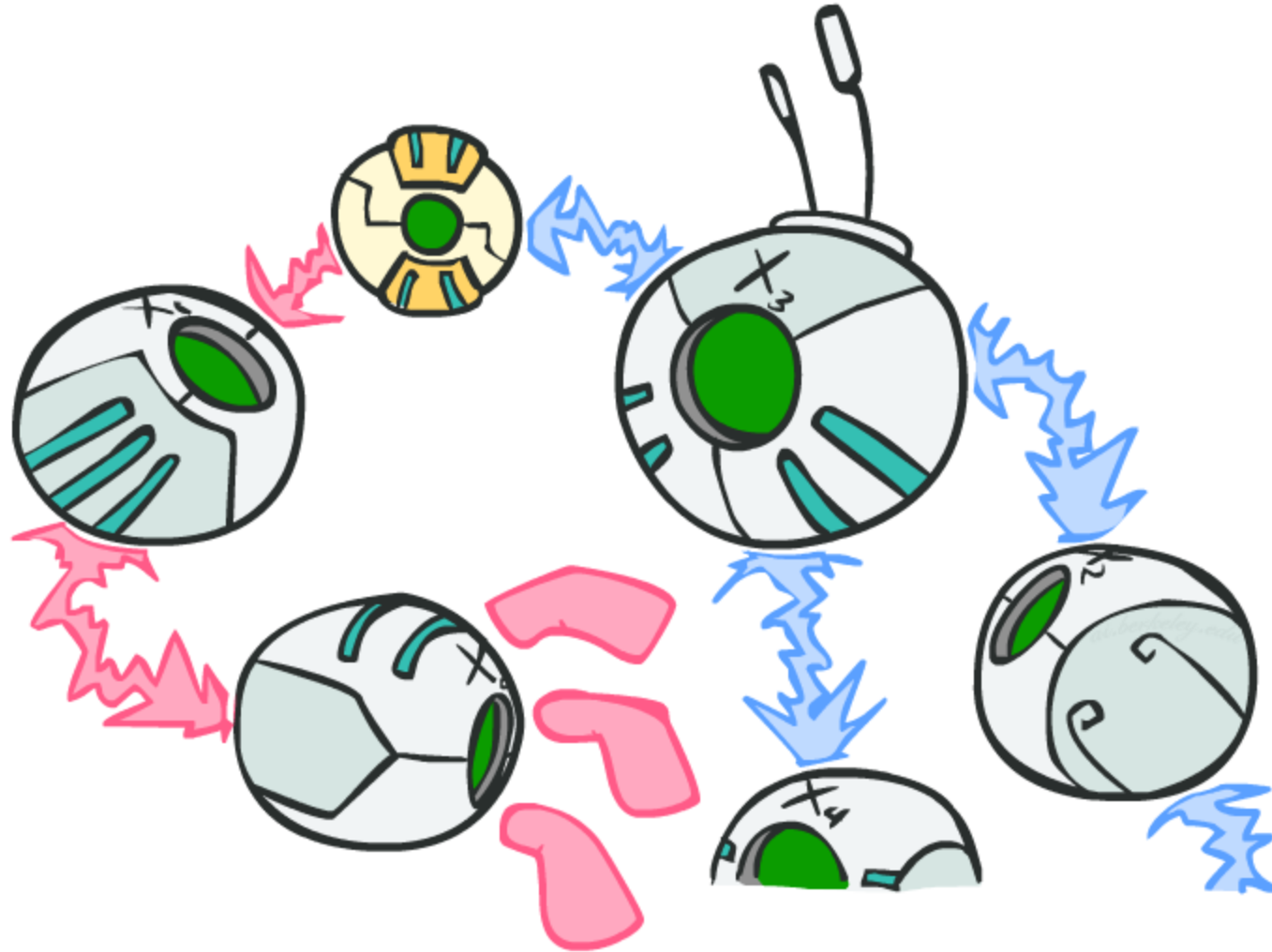
Example

- Burglary independent of Earthquake given Alarm?
 - NO!
 - Given that the alarm has sounded, both burglary and earthquake become more likely
 - But if we then learn that a burglary has happened, the alarm is **explained away** and the probability of earthquake drops back
- Burglary independent of Earthquake given JohnCalls?
- Any simple algorithm to determine conditional independence?

V-structure



D-separation



Causal Chains

- This configuration is a “causal chain”



X: Low pressure

Y: Rain

Z: Traffic

Global semantics:

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- Guaranteed X independent of Z ? **No!**
- Guaranteed X independent of Z given Y?

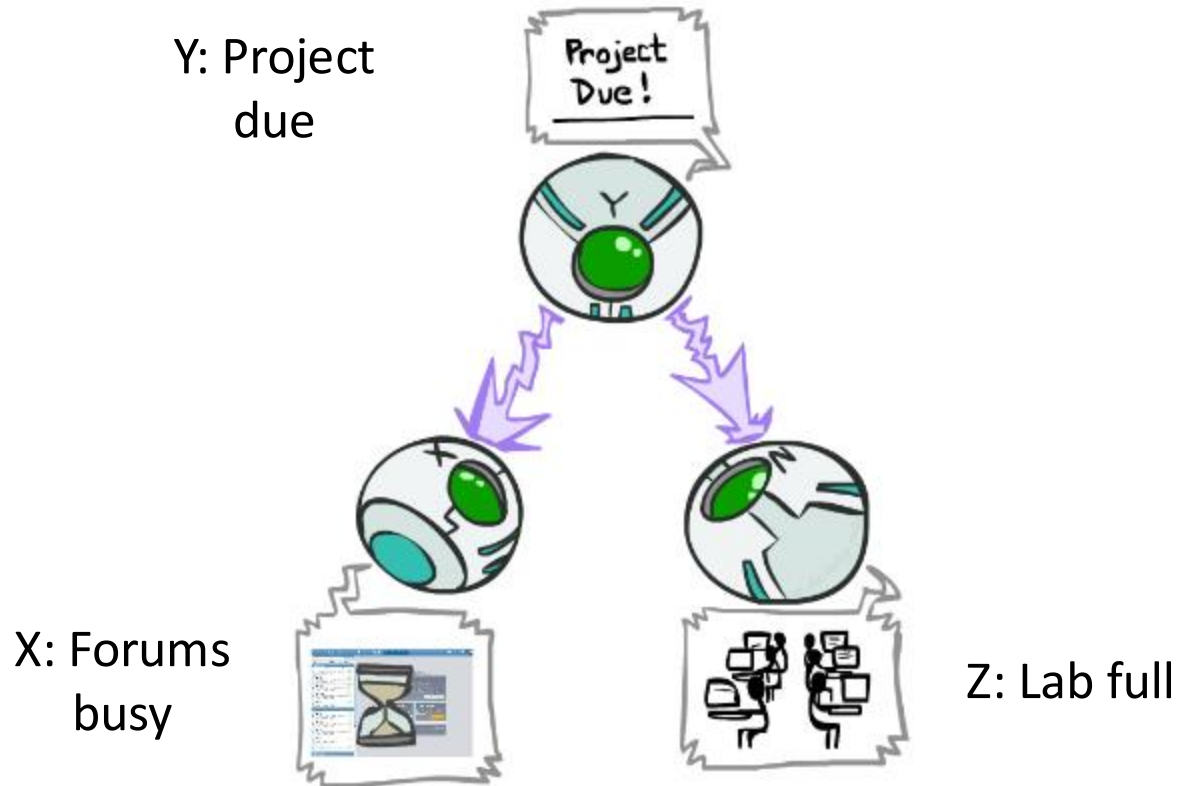
$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)} \\ &= P(z|y) \end{aligned}$$

Yes!

- Evidence along the chain “blocks” the influence

Common Cause

- This configuration is a “common cause”



Global semantics:

$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

- Guaranteed X independent of Z ? **No!**
- Guaranteed X and Z independent given Y?

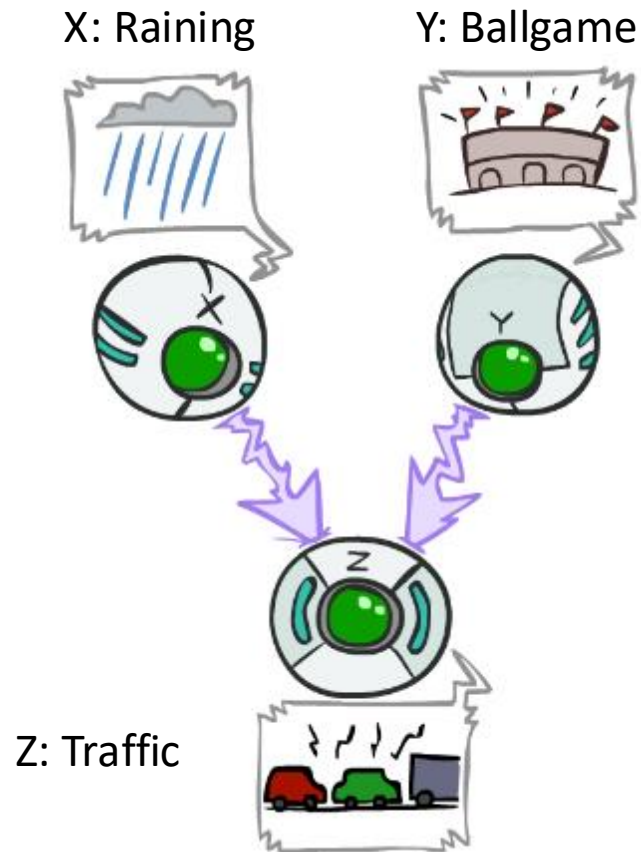
$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)} \\ &= P(z|y) \end{aligned}$$

Yes!

- Observing the cause blocks influence between effects.

Common Effect

- Last configuration: two causes of one effect (v-structures)



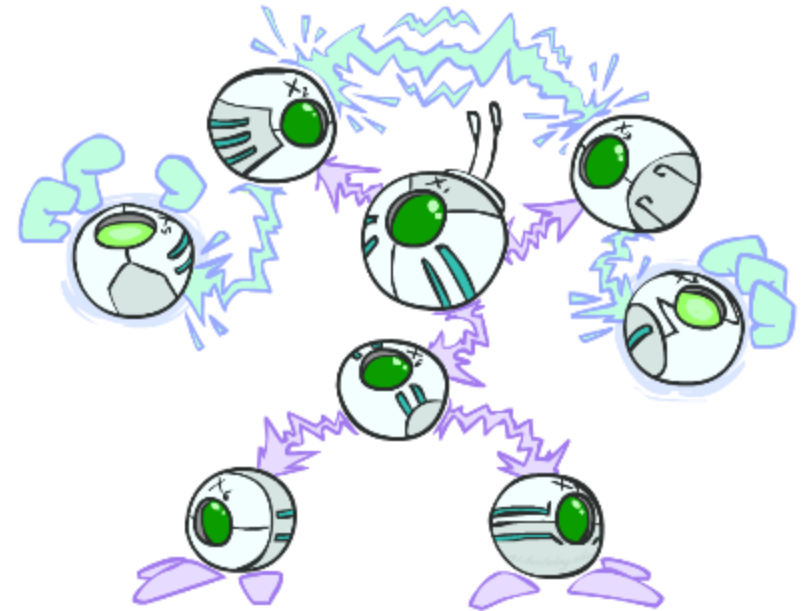
- Are X and Y independent?
 - **Yes**: the ballgame and the rain cause traffic, but they are not correlated
 - Still need to prove they must be (try it!)
- Are X and Y independent given Z?
 - **No**: seeing traffic puts the rain and the ballgame in competition as explanation.
- This is backwards from the other cases
 - Observing an effect **activates** influence between possible causes.

D-separation - the General Case



D-separation - the General Case

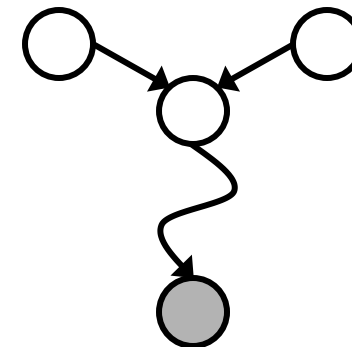
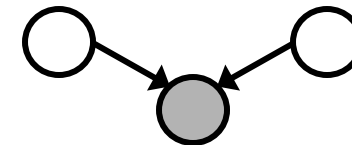
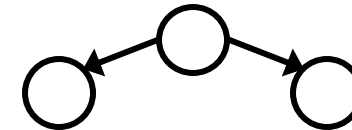
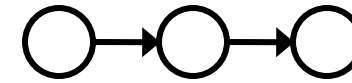
- General question: in a given BN, are two variables independent (given evidence)?
- Solution: analyze the graph; break the question into repetitions of the three canonical cases



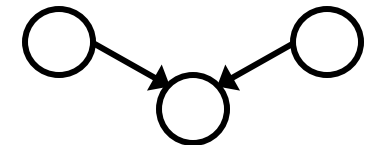
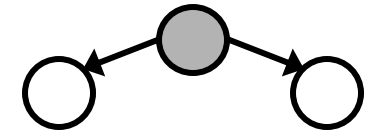
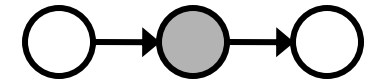
Active / Inactive Paths

- Question: X, Y, Z are non-intersecting subsets of nodes. Are X and Y conditionally independent given Z ?
- A triple is active in the following three cases
 - Causal chain $A \rightarrow B \rightarrow C$ where B is unobserved (either direction)
 - Common cause $A \leftarrow B \rightarrow C$ where B is unobserved
 - Common effect (aka v-structure)
 $A \rightarrow B \leftarrow C$ where B or one of its descendants is observed
- A path is active if each triple along the path is active
- A path is blocked if it contains a single inactive triple
- If all paths from X to Y are blocked, then X is said to be “**d-separated**” from Y by Z
- If d-separated, then X and Y are conditionally independent given Z

Active Triples



Inactive Triples



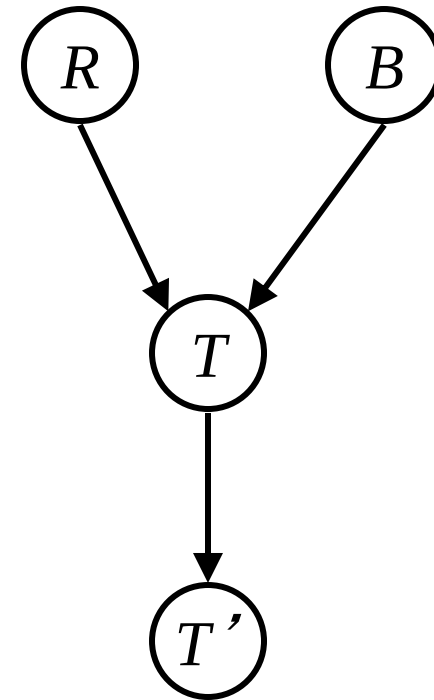
Example

$$R \perp\!\!\!\perp B$$

Yes

$$R \perp\!\!\!\perp B | T$$

$$R \perp\!\!\!\perp B | T'$$



Example

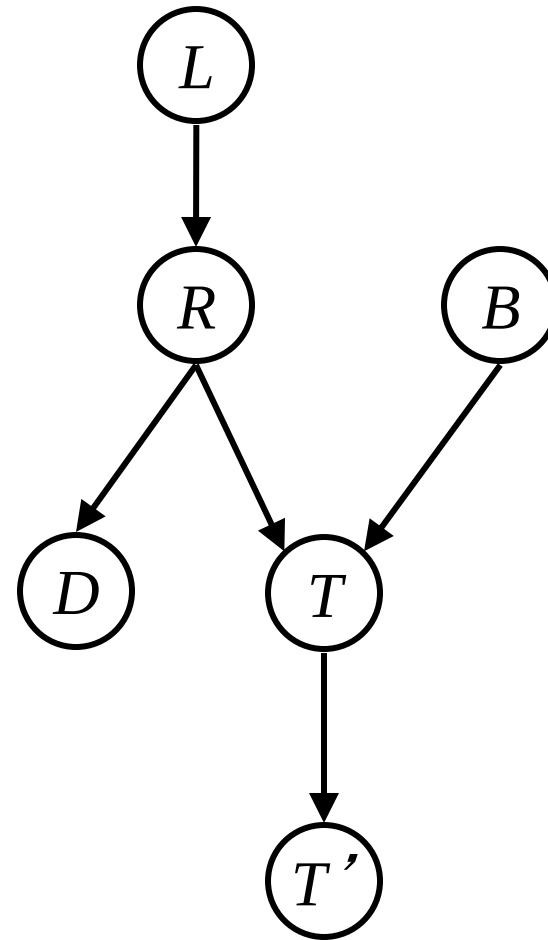
$L \perp\!\!\!\perp T' | T$ *Yes*

$L \perp\!\!\!\perp B$ *Yes*

$L \perp\!\!\!\perp B | T$

$L \perp\!\!\!\perp B | T'$

$L \perp\!\!\!\perp B | T, R$ *Yes*



Example

- Variables:

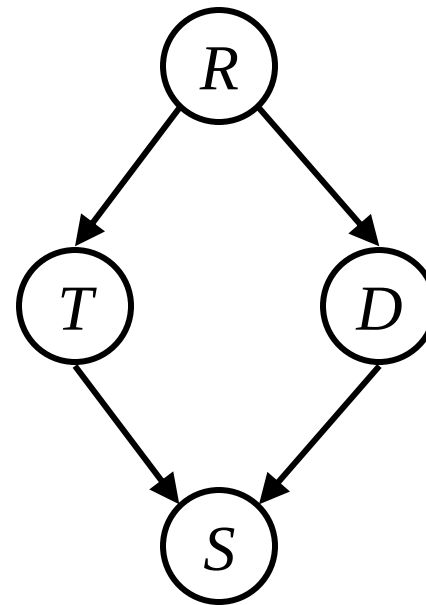
- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

- Questions:

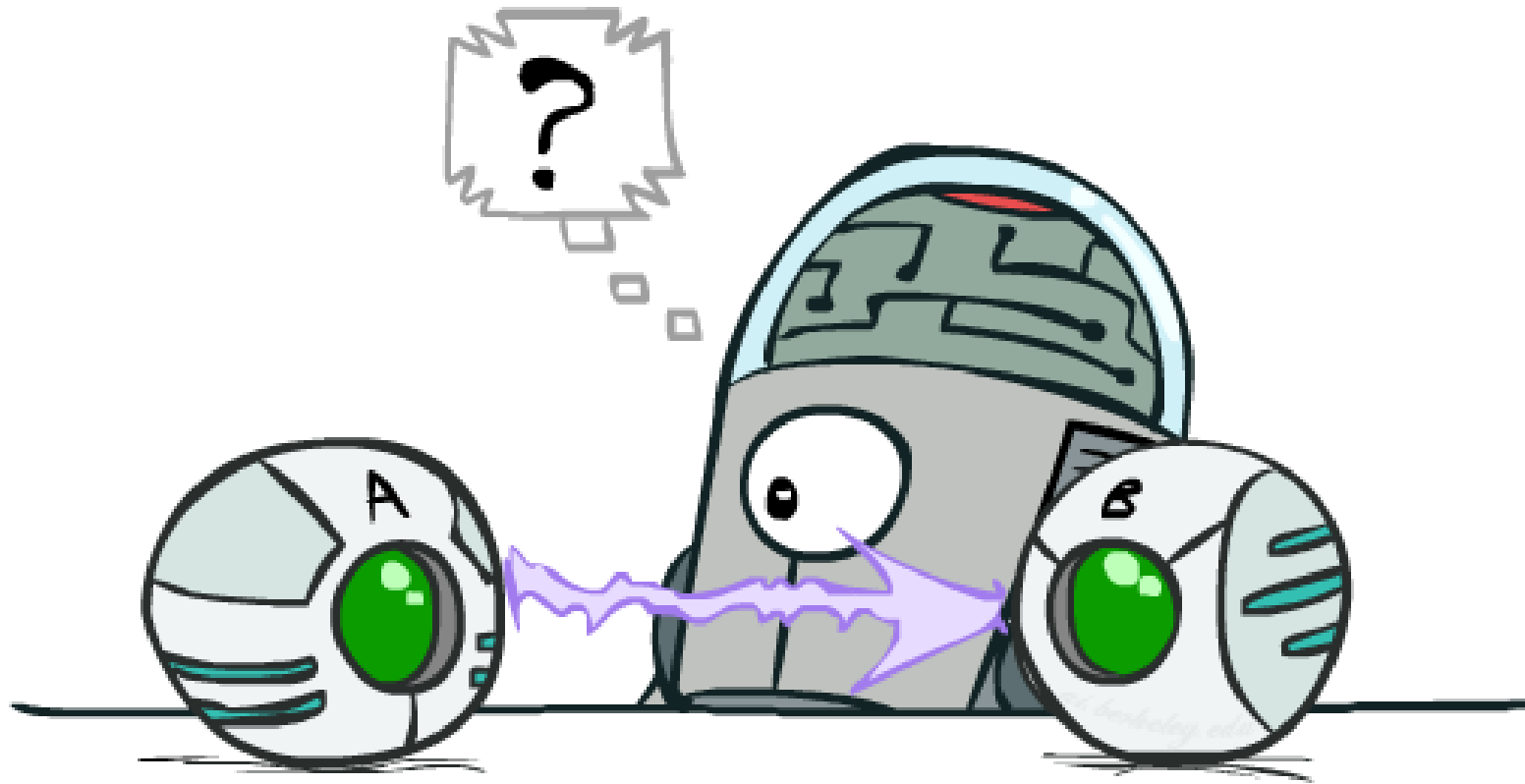
$$T \perp\!\!\!\perp D$$

$$T \perp\!\!\!\perp D | R \quad \text{Yes}$$

$$T \perp\!\!\!\perp D | R, S$$

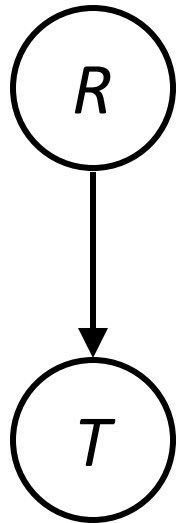


Node Ordering



Example: Traffic

- Causal direction



$P(R)$

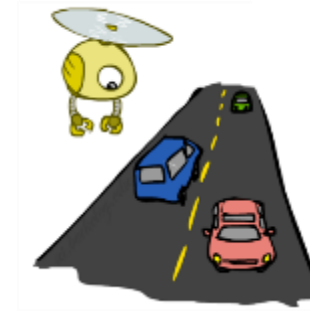
+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

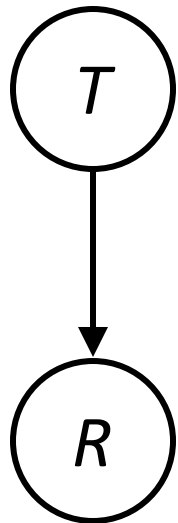
$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16



Example: Reverse Traffic

- Reverse causality?



$P(T)$

+t	9/16
-t	7/16

$P(R|T)$

+t	+r	1/3
	-r	2/3

-t	+r	1/7
	-r	6/7



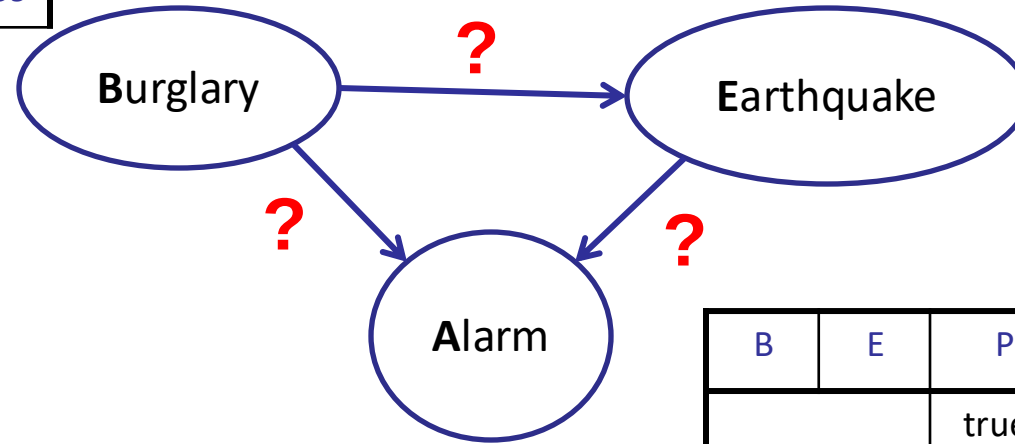
$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

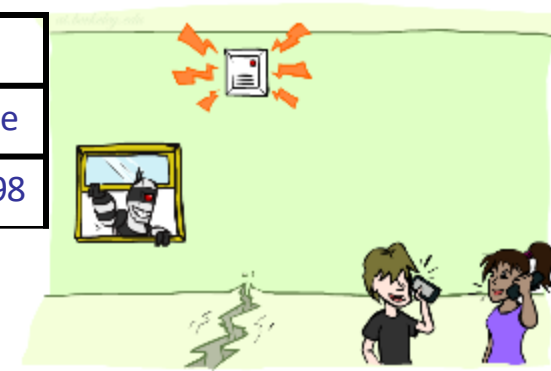
Example: Burglary

- Burglary
- Earthquake
- Alarm

P(B)	
true	false
0.001	0.999



P(E)	
true	false
0.002	0.998



B	E	P(A B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

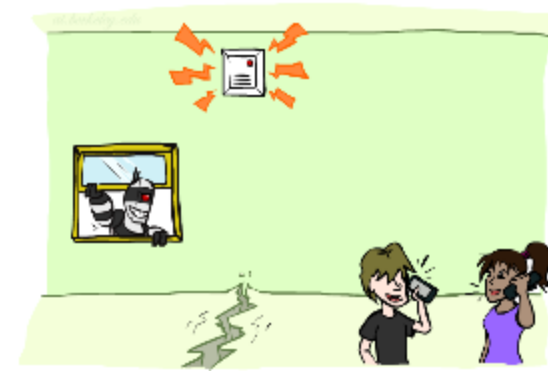
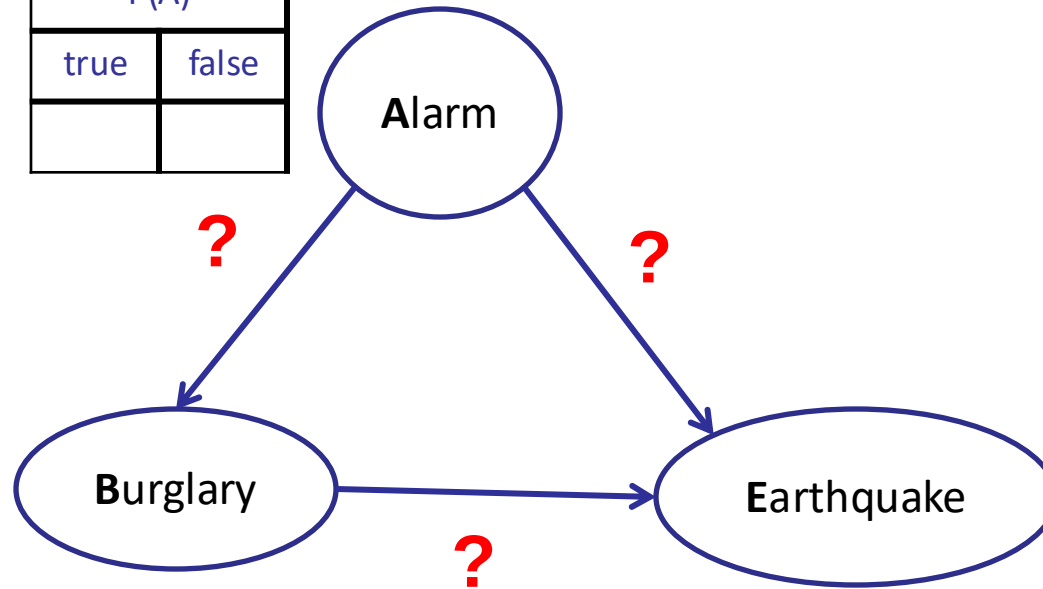
2 edges, 6 free parameters

Example: Burglary

- Alarm
- Burglary
- Earthquake

A	P(B A)	
	true	false
true		
false		

P(A)	
true	false

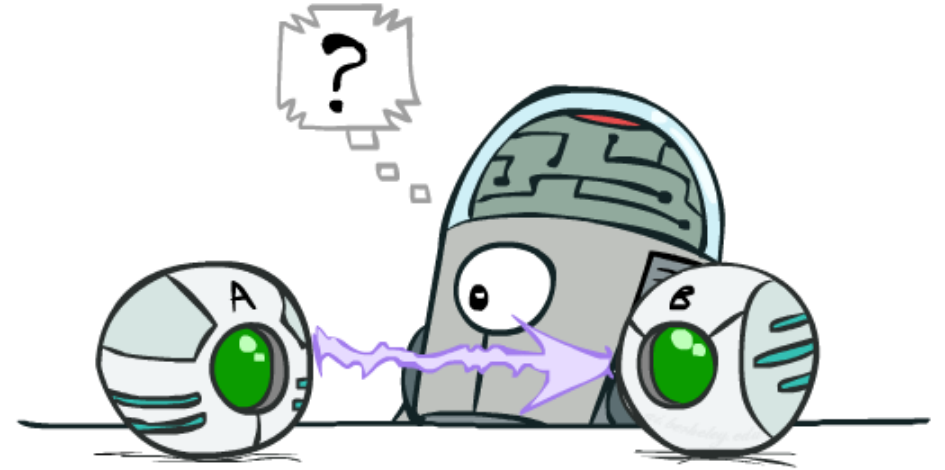


A	B	P(E A,B)	
		true	false
true	true		
true	false		
false	true		
false	false		

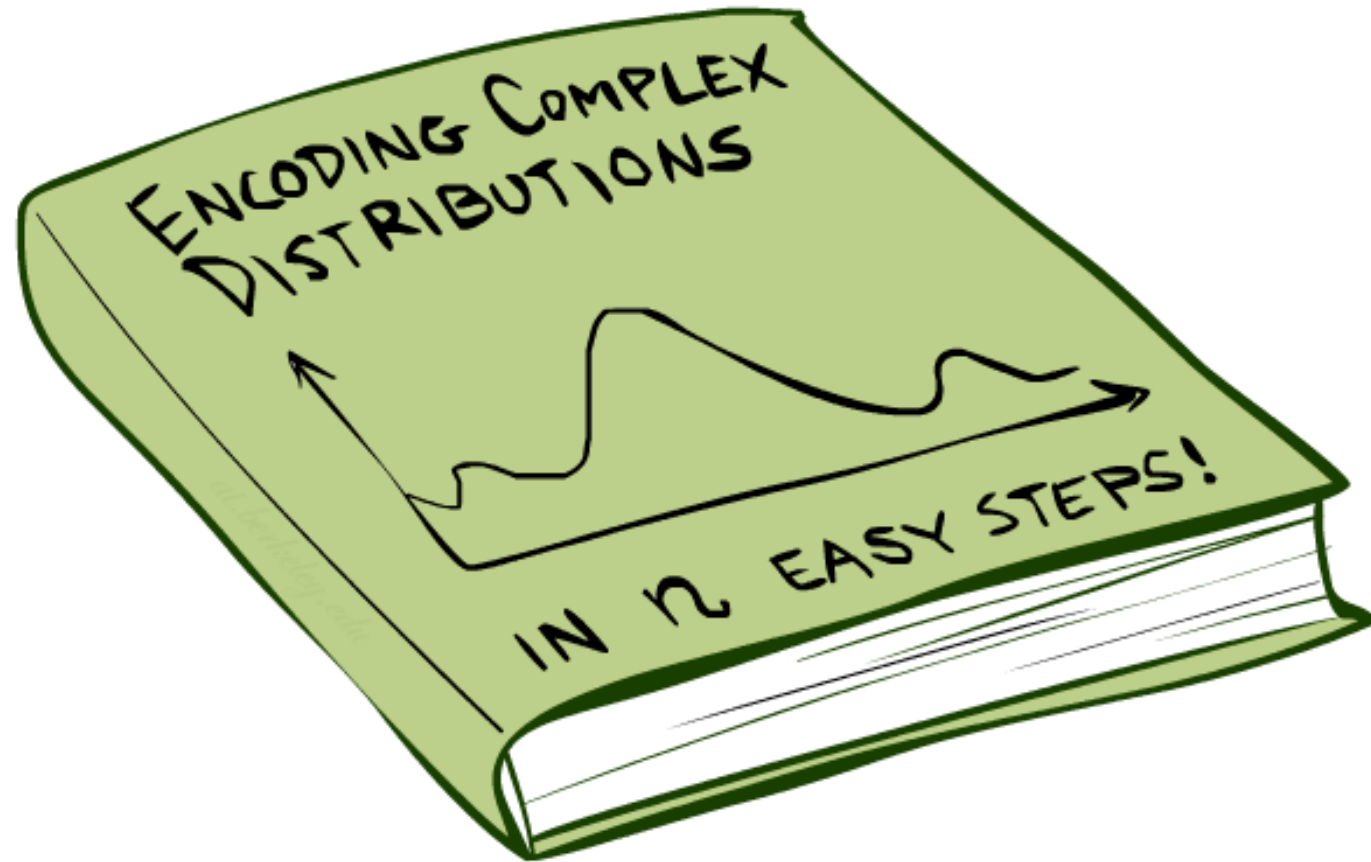
3 edges, 7 free parameters

Causality?

- When Bayes nets reflect the true causal patterns:
(e.g., Burglary, Earthquake, Alarm)
 - Often simpler (fewer parents, fewer parameters)
 - Often easier to assess probabilities
 - Often more robust: e.g., changes in frequency of burglaries should not affect the rest of the model!
- BNs need not actually be causal
 - Sometimes no causal net exists over the domain (especially if variables are missing)
 - E.g. consider the variables *Traffic* and *Umbrella*
 - End up with arrows that reflect correlation, not causation
- What do the arrows really mean?
 - Topology may happen to encode causal structure
 - **Topology really encodes conditional independence:**
 $P(X_i \mid X_1, \dots, X_{i-1}) = P(X_i \mid \text{Parents}(X_i))$



Markov Networks



Markov Networks

- A Bayesian network encodes a joint distribution with a directed acyclic graph
 - A CPT captures uncertainty between a node and its parents
- A Markov network (or Markov random field) encodes a joint distribution with an undirected graph
 - A potential function captures uncertainty between a clique of nodes

Markov Networks

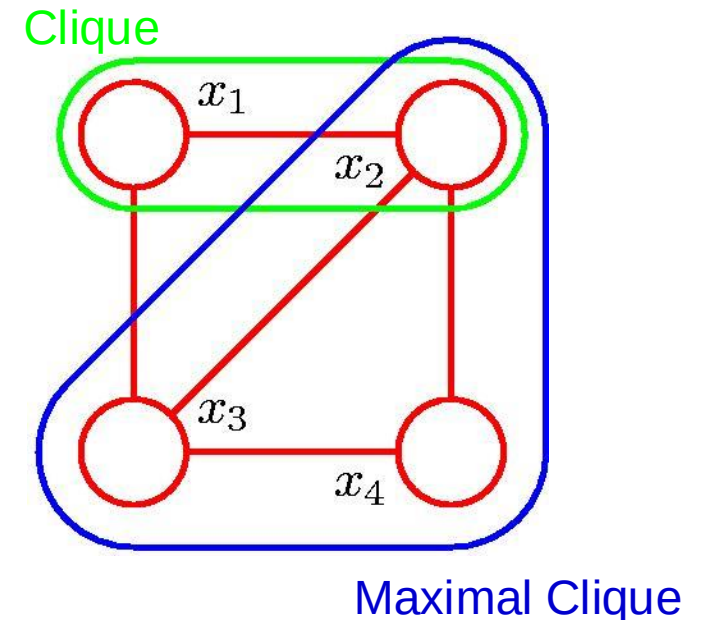
- Markov network = undirected graph + potential functions
 - For each clique (or max clique), a potential function is defined
 - A potential function is not locally normalized, i.e., it doesn't encode probabilities
 - A joint probability is proportional to the product of potentials

$$p(\mathbf{x}) = \frac{1}{Z} \prod_C \psi_C(\mathbf{x}_C)$$

where $\psi_C(\mathbf{x}_C)$ is the **potential** over **clique** C and

$$Z = \sum_{\mathbf{x}} \prod_C \psi_C(\mathbf{x}_C)$$

is the **normalization coefficient** (aka. partition function).



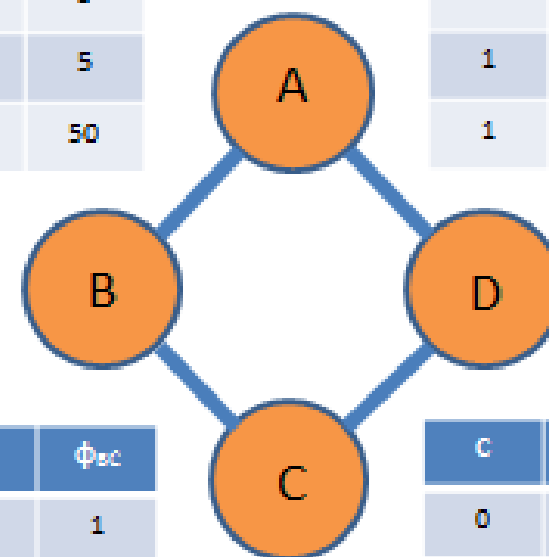
Markov Networks

A	B	C	D	$\phi_{AB}\phi_{BC}\phi_{CD}\phi_{AD}$
0	0	0	0	250
0	0	0	1	37500
0	0	1	0	50000
0	0	1	1	625000
0	1	0	0	1125
0	1	0	1	168750
0	1	1	0	50000
0	1	1	1	625000
1	0	0	0	250
1	0	0	1	375
1	0	1	0	50000
1	0	1	1	6250
1	1	0	0	112500
1	1	0	1	168750
1	1	1	0	5000000
1	1	1	1	625000

$$Z = 7520750$$

A	B	ϕ_{AB}
0	0	50
0	1	5
1	0	5
1	1	50

A	D	ϕ_{AD}
0	0	5
0	1	50
1	0	50
1	1	5

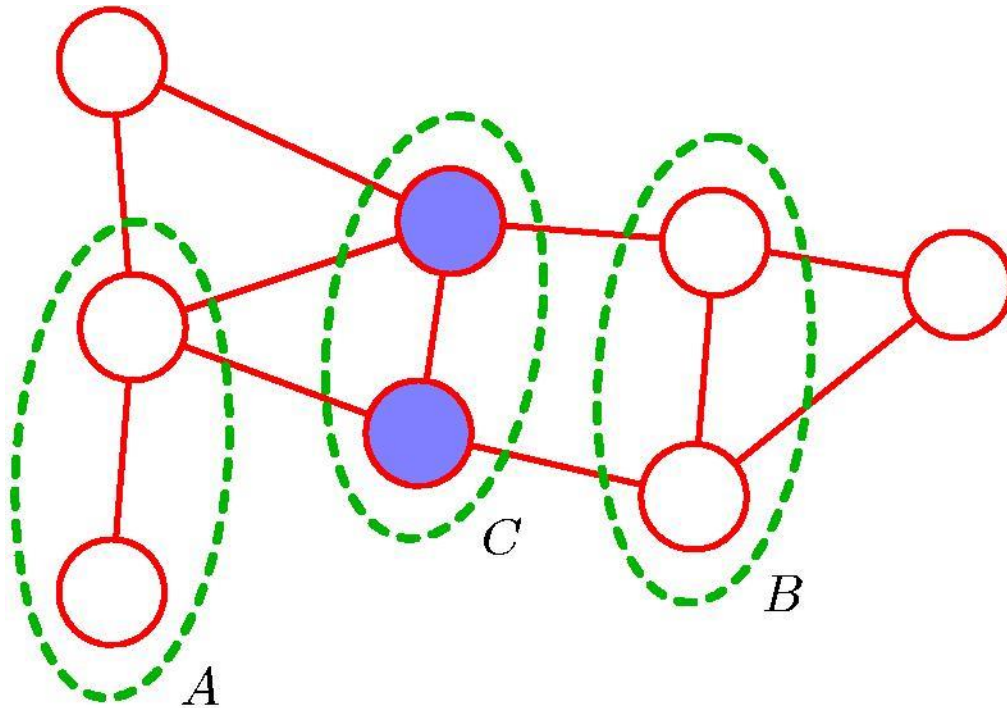


B	C	ϕ_{BC}
0	0	1
0	1	5
1	0	45
1	1	50

C	D	ϕ_{CD}
0	0	1
0	1	15
1	0	40
1	1	50

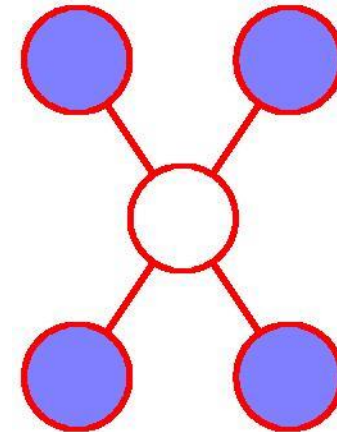
Markov Networks

- Conditional independence and Markov blanket in MN



$$A \perp\!\!\!\perp B | C$$

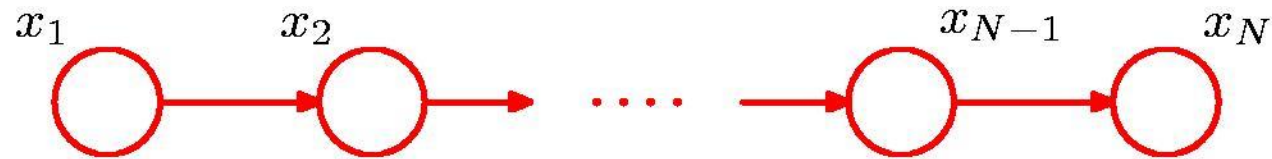
Markov Blanket



Graphical Models

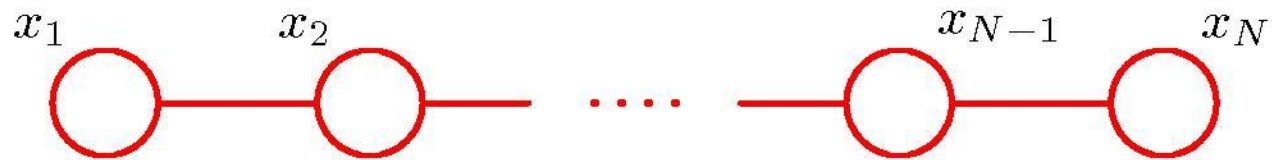
- A **graphical model** is a probabilistic model for which a graph expresses conditional dependence between random variables
 - Bayesian networks: directed acyclic graph
 - Markov networks: undirected graph
 - Factor graphs, conditional random fields, etc.

Converting Directed to Undirected Graphs (1)



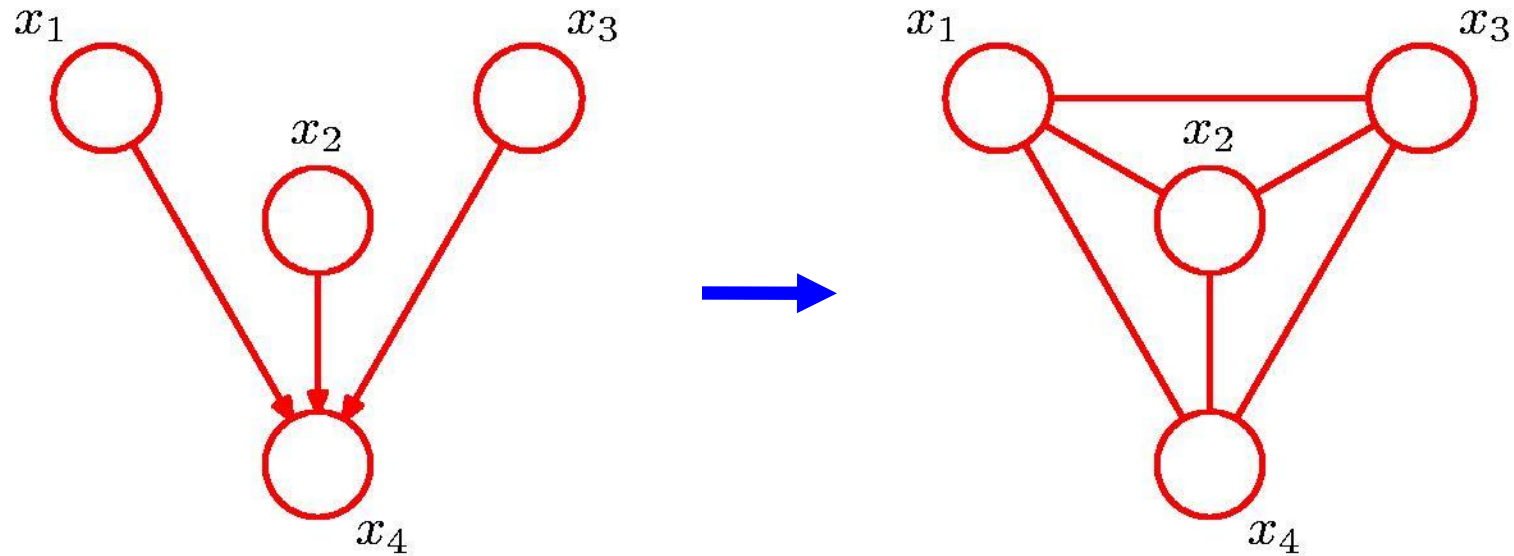
$$p(\mathbf{x}) = p(x_1) \underbrace{p(x_2|x_1)} \underbrace{p(x_3|x_2)} \cdots \underbrace{p(x_N|x_{N-1})}$$

$$p(\mathbf{x}) = \frac{1}{Z} \psi_{1,2}(x_1, x_2) \psi_{2,3}(x_2, x_3) \cdots \psi_{N-1,N}(x_{N-1}, x_N)$$



Converting Directed to Undirected Graphs (2)

- Additional links (moralization)

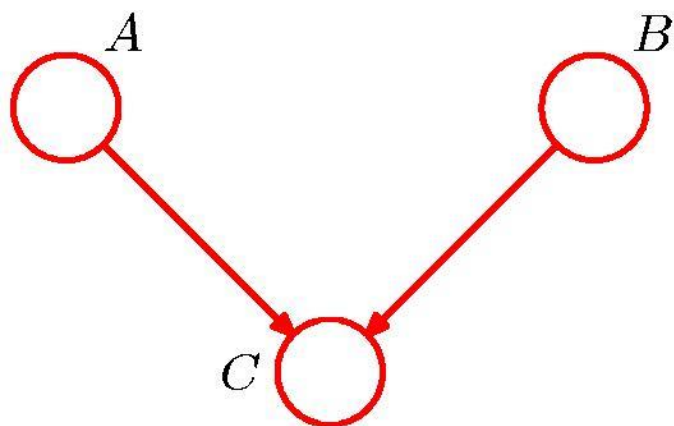


$$\begin{aligned} p(\mathbf{x}) &= p(x_1)p(x_2)p(x_3)p(x_4|x_1, x_2, x_3) \\ &= \frac{1}{Z} \psi(x_1, x_2, x_3, x_4) \end{aligned}$$

Bayesian Network $\rightarrow$ Markov Network

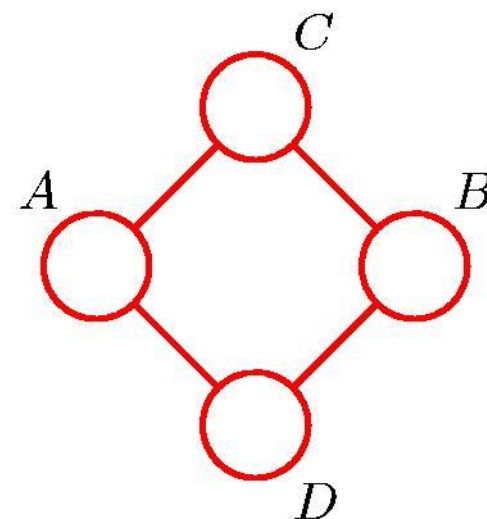
- Steps
 1. Moralization
 2. Construct potential functions from CPTs
- The BN and MN encode the same distribution
- Do they encode the same set of conditional independence?

Encoding Conditional Independence



$$A \perp\!\!\!\perp B \mid \emptyset$$

$$A \not\perp\!\!\!\perp B \mid C$$



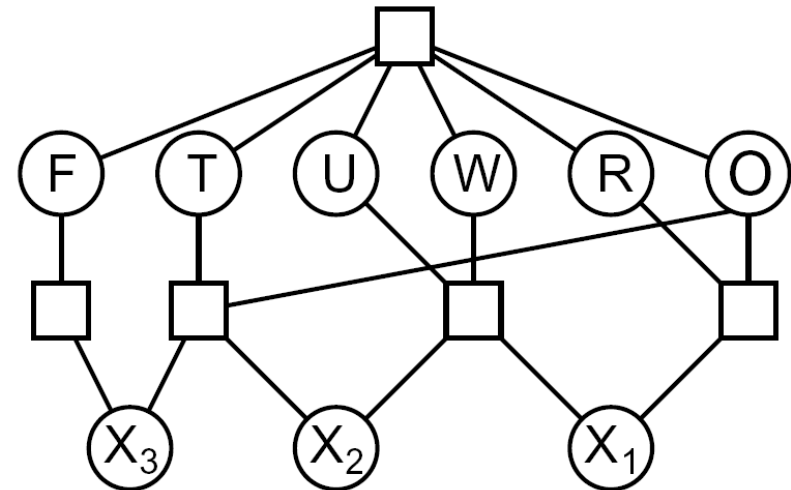
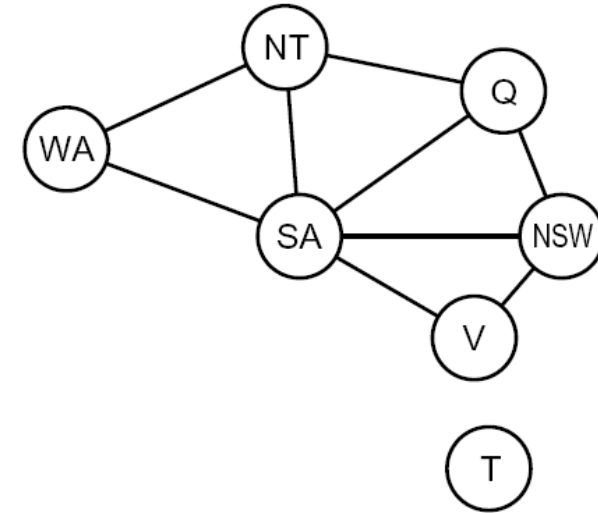
$$A \not\perp\!\!\!\perp B \mid \emptyset$$

$$A \perp\!\!\!\perp B \mid C \cup D$$

$$C \perp\!\!\!\perp D \mid A \cup B$$

Markov networks vs. Constraint graphs

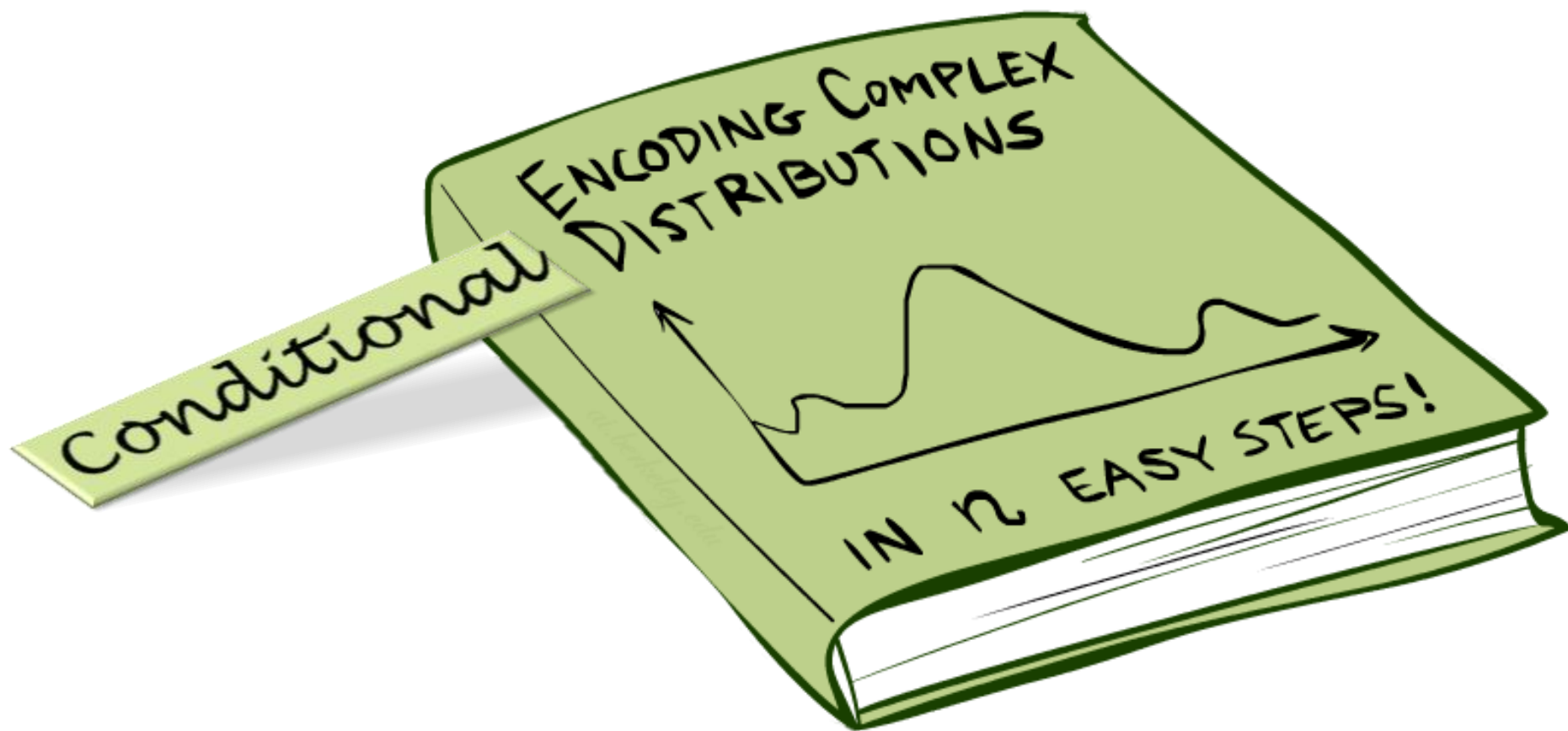
- Constraint graphs can be seen as Markov networks with 0/1 potentials



BN/MN vs. Logic

- Which logic is BN/MN more similar to: PL? FOL?
 - Boolean nodes represent propositions
 - No explicit representation of objects, relations, quantifiers
- BN/MN can be seen as a probabilistic extension of PL
- PL can be seen as BN/MN with deterministic CPTs/potentials

Conditional Random Fields



Generative vs. Discriminative Models

- Generative models

- A generative model represents a joint distribution $P(X_1, X_2, \dots, X_n)$
- Both BN and MN are generative models

- Discriminative models

- In some scenarios, we only care about predicting queries from evidence
- A discriminative model represents a conditional distribution $P(Y_1, Y_2, \dots, Y_n | X)$
- It does not model $P(X)$

Conditional Random Fields (CRF)

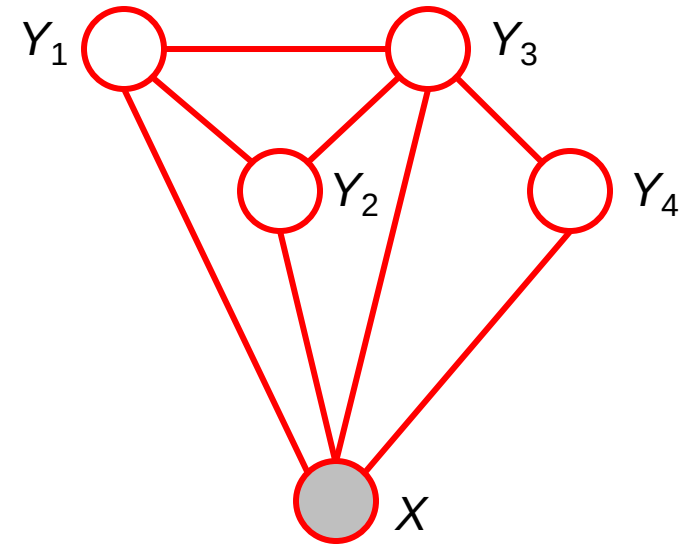
- An extension of MN (aka. Markov random field) where everything is conditioned on an input

$$P(y|x) = \frac{1}{Z(x)} \prod_c \psi_c(y_c, x)$$

where $\psi_c(y_c, x)$ is the potential over clique C and

$$Z(x) = \sum_y \prod_c \psi_c(y_c, x)$$

is the normalization coefficient.



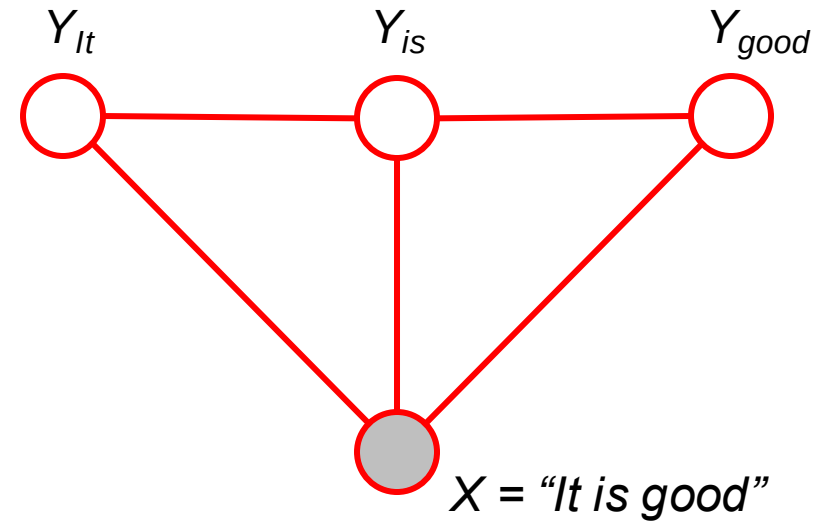
CRF Applications

- NLP

- POS tagging
- Named entity recognition
- Syntactic parsing

- CV

- Image segmentation
- Posture recognition



Summary

- A Bayesian network encodes a joint distribution
 - Syntax: DAG+CPTs
 - Semantics
 - Global semantics
 - Conditional independence semantics
 - D-separation
- Markov networks
 - Syntax: undirected graph + potentials
 - Semantics
 - Extension: CRF

